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Sora: A Review on Background, Technology, Limitations, and Opportunities of Large Vision Models

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Abstract

Note: This is not an official technical report from OpenAI.

Sora is a text-to-video generative AI model, released by OpenAI in February 2024. The model is trained to generate videos of realistic or imaginative scenes from text instructions and show potential in simulating the physical world. Based on public technical reports and reverse engineering, this paper presents a comprehensive review of the model's background, related technologies, applications, remaining challenges, and future directions of text-to-video AI models. We first trace Sora's development and investigate the underlying technologies used to build this "world simulator". Then, we describe in detail the applications and potential impact of Sora in multiple industries ranging from film-making and education to marketing. We discuss the main challenges and limitations that need to be addressed to widely deploy Sora, such as ensuring safe and unbiased video generation. Lastly, we discuss the future development of Sora and video generation models in general, and how advancements in the field could enable new ways of human-AI interaction, boosting productivity and creativity of video generation.



Figure 1: Sora: A Breakthrough in AI-Powered Vision Generation.

*Equal contributions. The order was determined by rolling dice. Chujie, Ruoxi, Yuan, Yue, and Zhengqing are visiting students in the LAIR lab at Lehigh University. The GitHub link is https://github.com/lichao-sun/SoraReview

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Sora：大型视觉模型背景、技术、局限与机遇综述

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摘要

注意：这并非来自OpenAI的官方技术报告。

Sora is a text-to-video generative AI model, released by OpenAI in February 2024. The model is trained to generate videos of realistic or imaginative scenes from text instructions and show potential in simulating the physical world. Based on public technical reports and reverse engineering, this paper presents a comprehensive review of the model's background, related technologies, applications, remaining challenges, and future directions of text-to-video AI models. We first trace Sora's development and investigate the underlying technologies used to build this "world simulator". Then, we describe in detail the applications and potential impact of Sora in multiple industries ranging from film-making and education to marketing. We discuss the main challenges and limitations that need to be addressed to widely deploy Sora, such as ensuring safe and unbiased video generation. Lastly, we discuss the future development of Sora and video generation models in general, and how advancements in the field could enable new ways of human-AI interaction, boosting productivity and creativity of video generation.



图1：Sora——人工智能技术驱动的视觉生成领域的重大突破。

*各位的贡献程度相同，具体顺序是通过掷骰子决定的。楚杰、若曦、远、月以及正清都是正在莱海大学LAIR实验室做访问研究的学者。相关GitHub链接为https://github.com/lichao-sun/SoraReview

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1 Introduction

Since the release of ChatGPT in November 2022, the advent of AI technologies has marked a significant transformation, reshaping interactions and integrating deeply into various facets of daily life and industry [1, 2]. Building on this momentum, OpenAI released, in February 2024, `Sora`, a text-to-video generative AI model that can generate videos of realistic or imaginative scenes from text prompts. Compared to previous video generation models, `Sora` is distinguished by its ability to produce up to 1-minute long videos with high quality while maintaining adherence to user's text instructions [3]. This progression of `Sora` is the embodiment of the long-standing AI research mission of equipping AI systems (or AI Agents) with the capability of understanding and interacting with the physical world in motion. This involves developing AI models that are capable of not only interpreting complex user instructions but also applying this understanding to solve real-world problems through dynamic and contextually rich simulations.

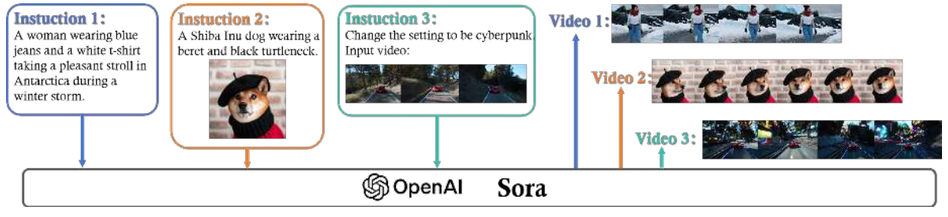


Figure 2: Examples of `Sora` in text-to-video generation. Text instructions are given to the OpenAI `Sora` model, and it generates three videos according to the instructions.

`Sora` demonstrates a remarkable ability to accurately interpret and execute complex human instructions, as illustrated in Figure 2. The model can generate detailed scenes that include multiple characters that perform specific actions against intricate backgrounds. Researchers attribute `Sora`'s proficiency to not only processing user-generated textual prompts but also discerning the complicated interplay of elements within a scenario. One of the most striking aspects of `Sora` is its capacity for up to a minute-long video while maintaining high visual quality and compelling visual coherency. Unlike earlier models that can only generate short video clips, `Sora`'s minute-long video creation possesses a sense of progression and a visually consistent journey from its first frame to the last. In addition, `Sora`'s advancements are evident in its ability to produce extended video sequences with nuanced depictions of motion and interaction, overcoming the constraints of shorter clips and simpler visual renderings that characterized earlier video generation models. This capability represents a leap forward in AI-driven creative tools, allowing users to convert text narratives to rich visual stories. Overall, these advances show the potential of `Sora` as a *world simulator* to provide nuanced insights into the physical and contextual dynamics of the depicted scenes. [3].

Technology. At the heart of `Sora` is a pre-trained *diffusion transformer* [4]. Transformer models have proven scalable and effective for many natural language tasks. Similar to powerful large language models (LLMs) such as GPT-4, `Sora` can parse text and comprehend complex user instructions. To make video generation computationally efficient, `Sora` employs *spacetime latent patches* as its building blocks. Specifically, `Sora` compresses a raw input video into a latent spacetime representation. Then, a sequence of latent spacetime patches is extracted from the compressed video to encapsulate both the visual appearance and motion dynamics over brief intervals. These patches, analogous to word tokens in language models, provide `Sora` with detailed *visual phrases* to be used to construct videos. `Sora`'s text-to-video generation is performed by a diffusion transformer model. Starting with a frame filled with visual noise, the model iteratively denoises the image and introduces specific details according to the provided text prompt. In essence, the

1 引言

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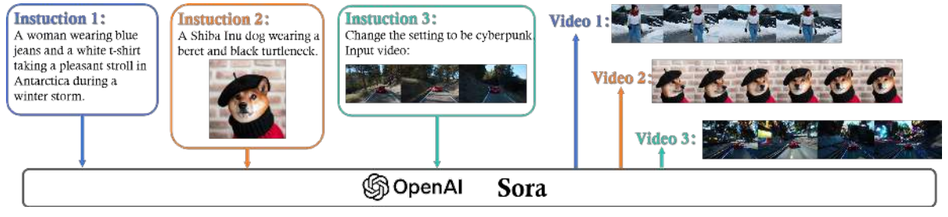


图2：Sora在文本转视频生成中的应用示例。系统向OpenAI的Sora模型输入文本指令，该模型便根据这些指令生成三段视频。

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generated video emerges through a multi-step refinement process, with each step refining the video to be more aligned with the desired content and quality.

Highlights of `Sora`. `Sora`'s capabilities have profound implications in various aspects:

- *Improving simulation abilities*: Training `Sora` at scale is attributed to its remarkable ability to simulate various aspects of the physical world. Despite lacking explicit 3D modeling, `Sora` exhibits 3D consistency with dynamic camera motion and long-range coherence that includes object persistence and simulates simple interactions with the world. Moreover, `Sora` intriguingly simulates digital environments like Minecraft, controlled by a basic policy while maintaining visual fidelity. These emergent abilities suggest that scaling video models is effective in creating AI models to simulate the complexity of physical and digital worlds.
- *Boosting creativity*: Imagine outlining a concept through text, whether a simple object or a full scene, and seeing a realistic or highly stylized video rendered within seconds. `Sora` allows an accelerated design process for faster exploration and refinement of ideas, thus significantly boosting the creativity of artists, filmmakers, and designers.
- *Driving educational innovations*: Visual aids have long been integral to understanding important concepts in education. With `Sora`, educators can easily turn a class plan from text to videos to captivate students' attention and improve learning efficiency. From scientific simulations to historical dramatizations, the possibilities are boundless.
- *Enhancing Accessibility*: Enhancing accessibility in the visual domain is paramount. `Sora` offers an innovative solution by converting textual descriptions to visual content. This capability empowers all individuals, including those with visual impairments, to actively engage in content creation and interact with others in more effective ways. Consequently, it allows for a more inclusive environment where everyone has the opportunity to express his or her ideas through videos.
- *Fostering emerging applications*: The applications of `Sora` are vast. For example, marketers might use it to create dynamic advertisements tailored to specific audience descriptions. Game developers might use it to generate customized visuals or even character actions from player narratives.

Limitations and Opportunities. While `Sora`'s achievements highlight significant advancements in AI, challenges remain. Depicting complex actions or capturing subtle facial expressions are among the areas where the model could be enhanced. In addition, ethical considerations such as mitigating biases in generated content and preventing harmful visual outputs underscore the importance of responsible usage by developers, researchers, and the broader community. Ensuring that `Sora`'s outputs are consistently safe and unbiased is a principal challenge. The field of video generation is advancing swiftly, with academic and industry research teams making relentless strides. The advent of competing text-to-video models suggests that `Sora` may soon be part of a dynamic ecosystem. This collaborative and competitive environment fosters innovation, leading to improved video quality and new applications that help improve the productivity of workers and make people's lives more entertaining.

Our Contributions. Based on published technical reports and our reverse engineering, this paper presents the first comprehensive review of `Sora`'s background, related technologies, emerging applications, current limitations, and future opportunities.

2 Background

2.1 History

In the realm of computer vision (CV), prior to the deep learning revolution, traditional image generation techniques relied on methods like texture synthesis [5] and texture mapping [6], based on hand-crafted features. However, these methods were limited in their capacity to produce complex and vivid images.

生成视频是通过多步骤优化流程制作而成的，每一步都会对视频进行精细调整，使其更贴合预期的内容和质量标准。

重点介绍`Sora`。 Sora的强大功能在多个领域都具有深远的意义：

- 提升模拟能力：大规模训练Sora的关键在于其出色的模拟物理世界各层面特性的能力。尽管缺乏明确的3D建模功能，Sora仍能展现出与动态摄像机运动相匹配的3D一致性，以及包含物体持续性在内的长距离连贯性，还能模拟与世界的简单交互。更为有趣的是，Sora能够在遵循基本策略控制的前提下，模拟《我的世界》这类数字环境，同时保持较高的视觉逼真度。这些涌现出的能力表明，扩展视频模型规模是打造能够模拟物理世界和数字世界复杂性的人工智能技术的有效途径。

- 提升创造力：试想通过文字来勾勒出一个概念，无论是简单的物体还是完整的场景，随后在几秒钟内就能看到渲染出的逼真或极具风格化的视频。Sora能够加速设计流程，帮助人们更快速地探索和完善创意，从而显著提升艺术家、电影制作者以及设计师的创造力。

- 推动教育创新：视觉辅助工具长期以来一直是帮助人们理解教育中重要概念的关键手段。借助Sora，教育工作者能够轻松将课堂计划从文字形式转化为视频，从而吸引学生的注意力并提升学习效率。无论是科学模拟还是历史场景再现，其应用可能性都是无穷的。

- 提升可访问性：在视觉领域提升可访问性至关重要。Sora通过将文本描述转换为视觉内容，提供了一种创新的解决方案。这一功能让包括视障人士在内的所有群体都能更积极地参与内容创作，并以更有效的方式与他人互动。因此，它营造了一个更加包容的环境，让每个人都有机会通过视频表达自己的想法。

- 推动新兴应用的发展：Sora的应用场景极为广泛。例如，营销人员可以利用它根据特定的目标受众特征制作动态广告；游戏开发者则可用其根据玩家的叙事内容生成定制化的视觉效果甚至角色动作。

局限性与机遇。 尽管Sora所取得的成就展现了人工智能技术的巨大进步，但依然存在诸多挑战。在描绘复杂动作或捕捉细微面部表情等方面，该模型仍有提升空间。此外，诸如减少生成内容中的偏见以及防止出现有害视觉输出等伦理问题，也凸显了开发者、研究人员以及更广泛群体负责任使用该技术的重要性。确保Sora的输出始终安全且无偏见是一项重大挑战。视频生成领域正在飞速发展，学术界和工业界的科研团队都在不断取得突破。随着各类竞争性文本转视频模型的出现，Sora很可能很快就会融入这个充满活力的生态系统之中。这种协作与竞争并存的氛围能够推动创新，进而提升视频质量，并带来新的应用，帮助提高工作效率，让人们的生活更加有趣。

我们的贡献。 基于已发表的技术报告以及我们进行的逆向工程，本文首次对Sora的背景、相关技术、新兴应用、当前局限以及未来发展机遇进行了全面梳理与分析。

2 背景

2.1 发展历程

在计算机视觉领域中，早在深度学习革命之前，传统的图像生成技术就依赖于基于手工设计特征的纹理合成 [5] 以及纹理映射 [6], 等方法。不过，这些方法在生成复杂且生动的图像方面存在诸多局限。

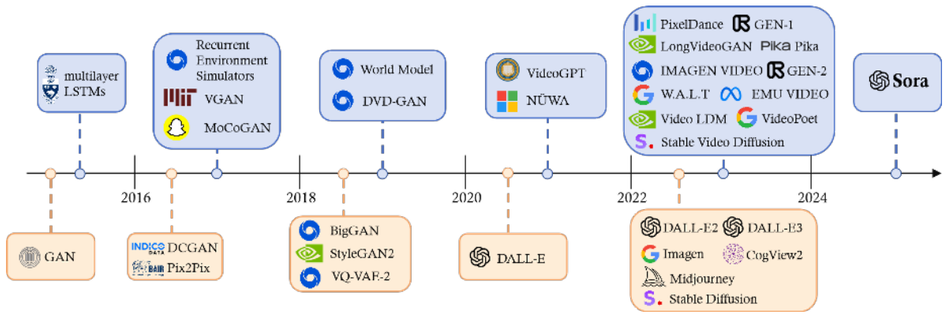


Figure 3: History of Generative AI in Vision Domain.

The introduction of Generative Adversarial Networks (GANs) [7] and Variational Autoencoders (VAEs) [8] marked a significant turning point due to its remarkable capabilities across various applications. Subsequent developments, such as flow models [9] and diffusion models [10], further enhanced image generation with greater detail and quality. The recent progress in Artificial Intelligence Generated Content (AIGC) technologies has democratized content creation, enabling users to generate desired content through simple textual instructions [11].

Over the past decade, the development of generative CV models has taken various routes, as shown in Figure 3. This landscape began to shift notably following the successful application of the transformer architecture [12] in NLP, as demonstrated by BERT [13] and GPT [14]. In CV, researchers take this concept even further by combining the transformer architecture with visual components, allowing it to be applied to downstream CV tasks, such as Vision Transformer (ViT) [15] and Swin Transformer [16]. Parallel to the transformer's success, diffusion models have also made significant strides in the fields of image and video generation [10]. Diffusion models offer a mathematically sound framework for converting noise into images with U-Nets [17], where U-Nets facilitate this process by learning to predict and mitigate noise at each step. Since 2021, a paramount focus in AI has been on generative language and vision models that are capable of interpreting human instructions, known as multimodal models. For example, CLIP [18] is a pioneering vision-language model that combines transformer architecture with visual elements, facilitating its training on vast datasets of text and images. By integrating visual and linguistic knowledge from the outset, CLIP can function as an image encoder within multimodal generation frameworks. Another notable example is Stable Diffusion [19], a versatile text-to-image AI model celebrated for its adaptability and ease of use. It employs transformer architecture and latent diffusion techniques to decode textual inputs and produce images of a wide array of styles, further illustrating the advancements in multimodal AI.

Following the release of ChatGPT in November 2022, we have witnessed the emergence of commercial text-to-image products in 2023, such as Stable Diffusion [19], Midjourney [20], DALL-E 3 [21]. These tools enable users to generate new images of high resolution and quality with simple text prompts, showcasing the potential of AI in creative image generation. However, transitioning from text-to-image to text-to-video is challenging due to the temporal complexity of videos. Despite numerous efforts in industry and academia, most existing video generation tools, such as Pika [22] and Gen-2 [23], are limited to producing only short video clips of a few seconds. In this context, `Sora` represents a significant breakthrough, akin to ChatGPT's impact in the NLP domain. `Sora` is the first model that is capable of generating videos up to one minute long based on human instructions, marking a milestone that profoundly influences research and development in generative AI. To facilitate easy access to the latest advancements in vision generation models, the most recent works have been compiled and provided in the Appendix and our GitHub.

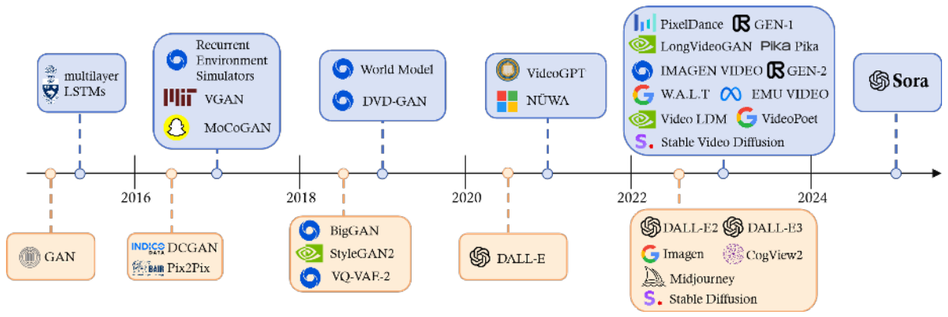


图3：视觉领域中生成式人工智能技术的发展历程。

生成对抗网络（GANs）的诞生 [7] 以及变分自编码器（VAEs）的出现 [8]为该领域带来了重大转折，因为它们在各种应用场景中都展现出了出色的性能。此后出现的流模型 [9] 和扩散模型 [10]，进一步提升了图像生成的细节程度与质量。而近期在人工智能技术驱动的内容生成领域取得的进展，则让内容创作变得更加普惠，用户只需通过简单的文本指令就能生成所需内容 [11]。

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Following the release of ChatGPT in November 2022, we have witnessed the emergence of commercial text-to-image products in 2023, such as Stable Diffusion [19], Midjourney [20], DALL-E 3 [21]. These tools enable users to generate new images of high resolution and quality with simple text prompts, showcasing the potential of AI in creative image generation. However, transitioning from text-to-image to text-to-video is challenging due to the temporal complexity of videos. Despite numerous efforts in industry and academia, most existing video generation tools, such as Pika [22] and Gen-2 [23], are limited to producing only short video clips of a few seconds. In this context, `Sora` represents a significant breakthrough, akin to ChatGPT's impact in the NLP domain. `Sora` is the first model that is capable of generating videos up to one minute long based on human instructions, marking a milestone that profoundly influences research and development in generative AI. To facilitate easy access to the latest advancements in vision generation models, the most recent works have been compiled and provided in the Appendix and our GitHub.

2.2 Advanced Concepts

Scaling Laws for Vision Models. With scaling laws for LLMs, it is natural to ask whether the development of vision models follows similar scaling laws. Recently, Zhai et al. [24] have demonstrated that the performance-compute frontier for ViT models with enough training data roughly follows a (saturating) power law. Following them, Google Research [25] presented a recipe for highly efficient and stable training of a 22B-parameter ViT. Results show that great performance can be achieved using the frozen model to produce embeddings, and then training thin layers on top. `Sora`, as a large vision model (LVM), aligns with these scaling principles, uncovering several emergent abilities in text-to-video generation. This significant progression underscores the potential for LVMs to achieve advancements like those seen in LLMs.

Emergent Abilities. Emergent abilities in LLMs are sophisticated behaviors or functions that manifest at certain scales—often linked to the size of the model's parameters—that were not explicitly programmed or anticipated by their developers. These abilities are termed "emergent" because they emerge from the model's comprehensive training across varied datasets, coupled with its extensive parameter count. This combination enables the model to form connections and draw inferences that surpass mere pattern recognition or rote memorization. Typically, the emergence of these abilities cannot be straightforwardly predicted by extrapolating from the performance of smaller-scale models. While numerous LLMs, such as ChatGPT and GPT-4, exhibit emergent abilities, vision models demonstrating comparable capabilities have been scarce until the advent of Sora. According to `Sora`'s technical report, it is the first vision model to exhibit confirmed emergent abilities, marking a significant milestone in the field of computer vision.

In addition to its emergent abilities, `Sora` exhibits other notable capabilities, including instruction following, visual prompt engineering, and video understanding. These aspects of `Sora`'s functionality represent significant advancements in the vision domain and will be explored and discussed in the rest sections.

3 Technology

3.1 Overview of `Sora`

In the core essence, `Sora` is a diffusion transformer [4] with flexible sampling dimensions as shown in Figure 4. It has three parts: (1) A time-space compressor first maps the original video into latent space. (2) A ViT then processes the tokenized latent representation and outputs the denoised latent representation. (3) A CLIP-like [26] conditioning mechanism receives LLM-augmented user instructions and potentially

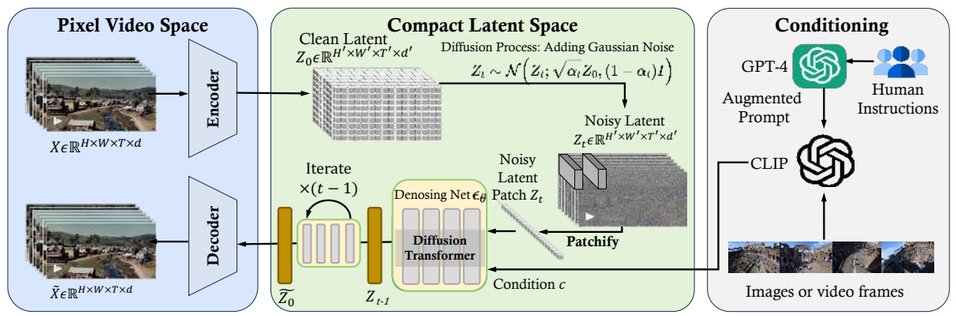


Figure 4: **Reverse Engineering**: Overview of `Sora` framework

2.2 高级概念

Scaling Laws for Vision Models. With scaling laws for LLMs, it is natural to ask whether the development of vision models follows similar scaling laws. Recently, Zhai et al. [24] have demonstrated that the performance-compute frontier for ViT models with enough training data roughly follows a (saturating) power law. Following them, Google Research [25] presented a recipe for highly efficient and stable training of a 22B-parameter ViT. Results show that great performance can be achieved using the frozen model to produce embeddings, and then training thin layers on top. `Sora`, as a large vision model (LVM), aligns with these scaling principles, uncovering several emergent abilities in text-to-video generation. This significant progression underscores the potential for LVMs to achieve advancements like those seen in LLMs.

涌现能力。大语言模型中的涌现能力指的是在特定规模下——通常与模型参数的数量相关——表现出来的复杂行为或功能，而这些是其开发者并未明确编程或预料的。这类能力被称为“涌现能力”，是因为它们源自模型在各类数据集上的全面训练以及庞大的参数量。这样的组合使得模型能够建立联系并做出推理，其能力远超单纯的模式识别或死记硬背。通常，无法通过从较小规模模型的性能推断来直接预测这些能力的出现。尽管诸如ChatGPT和GPT-4等众多大语言模型都展现出涌现能力，但在Sora出现之前，具备类似功能的视觉模型却极为罕见。根据Sora的技术报告，它是首个被证实具有涌现能力的视觉模型，为计算机视觉领域树立了重要的里程碑。

除了那些涌现出的能力之外，Sora还具备其他值得关注的功能，包括指令遵循、视觉提示工程以及视频理解能力。Sora在这些功能方面的表现代表了视觉领域内的重大进步，后续章节将对它们进行深入探讨与分析。

3 技术架构

3.1 Sora 概述

从核心本质来看，`Sora` 是一种扩散变换器 [4] ，其采样维度具有高度灵活性，如图4所示。它由三部分组成：(1) 时空压缩器，用于先将原始视频映射到潜在空间。
(2)随后，ViT模型会对分词后的潜在表示进行处理，并输出去噪后的潜在表示。*(3)类CLIP的 [26] 条件机制会接收经大语言模型增强过的用户指令，进而

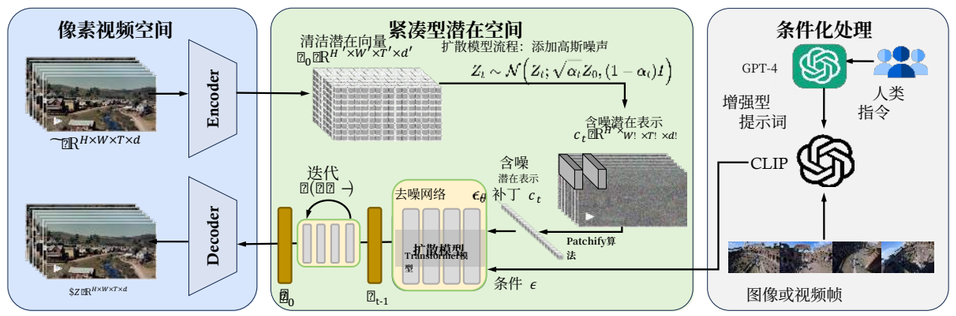


图4：**逆向工程**：Sora框架概览

visual prompts to guide the diffusion model to generate styled or themed videos. After many denoising steps, the latent representation of the generated video is obtained and then mapped back to pixel space with the corresponding decoder. In this section, we aim to reverse engineer the technology used by Sora and discuss a wide range of related works.

3.2 Data Pre-processing

3.2.1 Variable Durations, Resolutions, Aspect Ratios

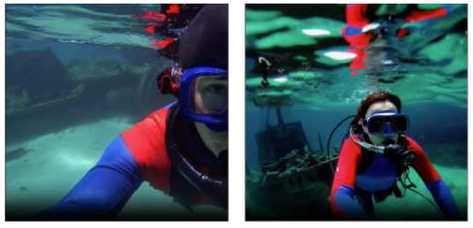
One distinguishing feature of Sora is its ability to train on, understand, and generate videos and images at their native sizes [3] as illustrated in Figure 5. Traditional methods often resize, crop, or adjust the aspect ratios of videos to fit a uniform standard—typically short clips with square frames at fixed low resolutions [27][28][29]. Those samples are often generated at a wider temporal stride and rely on separately trained frame-insertion and resolution-rendering models as the final step, creating inconsistency across the video. Utilizing the diffusion transformer architecture [4] (see Section 3.2.4), Sora is the first model to embrace the diversity of visual data and can sample in a wide array of video and image formats, ranging from widescreen 1920x1080p videos to vertical 1080x1920p videos and everything in between without compromising their original dimensions.



(a) Vertical (b) Square (c) Horizontal

Figure 5: Sora can generate images in flexible sizes or resolutions ranging from 1920x1080p to 1080x1920p and anything in between.

Training on data in their native sizes significantly improves composition and framing in the generated videos. Empirical findings suggest that by maintaining the original aspect ratios, Sora achieves a more natural and coherent visual narrative. The comparison between Sora and a model trained on uniformly cropped square videos demonstrates a clear advantage as shown in Figure 6. Videos produced by Sora exhibit better framing, ensuring subjects are fully captured in the scene, as opposed to the sometimes truncated views resulting from square cropping.



(a) Training on videos that are cropped to squares leads to unnatural compositions and framing.

(b) Training in native sizes improves framing.

Figure 6: A comparison between Sora (right) and a modified version of the model (left), which crops videos to square shapes—a common practice in model training—highlights the advantages.

This nuanced understanding and preservation of original video and image characteristics mark a significant advancement in the field of generative models. Sora's approach not only showcases the potential for more authentic and engaging video generation but also highlights the importance of diversity in

这些视觉提示用于引导扩散模型生成具有特定风格或主题的视频。经过多次去噪步骤后，即可得到所生成视频的潜在表示，再通过相应的解码器将其映射回像素空间。在本章节中，我们将对Sora所采用的技术进行逆向分析，并探讨各类相关研究。

3.2 数据预处理

3.2.1 不同的时长、分辨率与宽高比

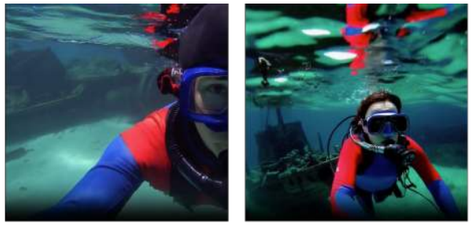
One distinguishing feature of Sora is its ability to train on, understand, and generate videos and images at their native sizes [3] as illustrated in Figure 5. Traditional methods often resize, crop, or adjust the aspect ratios of videos to fit a uniform standard—typically short clips with square frames at fixed low resolutions [27][28][29]. Those samples are often generated at a wider temporal stride and rely on separately trained frame-insertion and resolution-rendering models as the final step, creating inconsistency across the video. Utilizing the diffusion transformer architecture [4] (see Section 3.2.4), Sora is the first model to embrace the diversity of visual data and can sample in a wide array of video and image formats, ranging from widescreen 1920x1080p videos to vertical 1080x1920p videos and everything in between without compromising their original dimensions.



(a) Vertical (b) Square (c) Horizontal

图5：Sora能够生成尺寸或分辨率灵活的图像，范围从1920x1080p到1080x1920p，以及两者之间的任意尺寸。

使用原始尺寸的数据进行训练，能显著提升生成视频的构图与画面框架质量。实证研究显示，通过保持原有的宽高比，Sora能够呈现出更加自然且连贯的视觉叙事。将Sora 与在统一裁剪的正方形视频上训练得到的模型进行对比后，如图6所示，前者具有明显的优势。由Sora 生成的视频具备更优的构图效果，能够确保场景中的主体被完整呈现，而不会像正方形裁剪那样出现内容被截断的情况。



(a) Training on videos that are cropped to squares leads to unnatural compositions and framing.

(b) Training in native sizes improves framing.

图6：展示了Sora（右侧）与经过修改的模型版本（左侧）的对比，后者会将视频裁剪为正方形，这是模型训练中的一种常见做法，该图突显了Sora的优势。

对原始视频与图像特性的这种细致理解及精准保留，标志着生成模型领域的一项重大进展。Sora所采用的方法不仅展现了其潜力……更高的真实性潜力。既展现了极具吸引力且引人入胜的视频生成技术，同时也强调了多样性在其中的重要性

training data for achieving high-quality results in generative AI. The training approach of Sora aligns with the core tenet of Richard Sutton's THE BITTER LESSON[30], which states that leveraging computation over human-designed features leads to more effective and flexible AI systems. Just as the original design of diffusion transformers seeks simplicity and scalability [31], Sora's strategy of training on data at their native sizes eschews traditional AI reliance on human-derived abstractions, favoring instead a generalist method that scales with computational power. In the rest of this section, we try to reverse engineer the architecture design of Sora and discuss related technologies to achieve this amazing feature.

3.2.2 Unified Visual Representation

To effectively process diverse visual inputs including images and videos with varying durations, resolutions, and aspect ratios, a crucial approach involves transforming all forms of visual data into a unified representation, which facilitates the large-scale training of generative models. Specifically, Sora patchifies videos by initially compressing videos into a lower-dimensional latent space, followed by decomposing the representation into spacetime patches. However, Sora's technical report [3] merely presents a high-level idea, making reproduction challenging for the research community. In this section, we try to reverse-engineer the potential ingredients and technical pathways. Additionally, we will discuss viable alternatives that could replicate Sora's functionalities, drawing upon insights from existing literature.

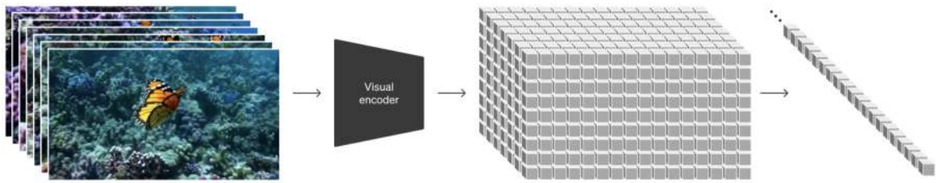


Figure 7: At a high level, Sora turns videos into patches by first compressing videos into a lower-dimensional latent space, and subsequently decomposing the representation into spacetime patches. Source: Sora's technical report [3].

3.2.3 Video Compression Network

Sora's video compression network (or visual encoder) aims to reduce the dimensionality of input data, especially a raw video, and output a latent representation that is compressed both temporally and spatially as shown in Figure 7. According to the references in the technical report, the compression network is built upon VAE or Vector Quantised-VAE (VQ-VAE) [32]. However, it is challenging for VAE to map visual data of any size to a unified and fixed-sized latent space if resizing and cropping are not used as mentioned in the technical report. We summarize two distinct implementations to address this issue:

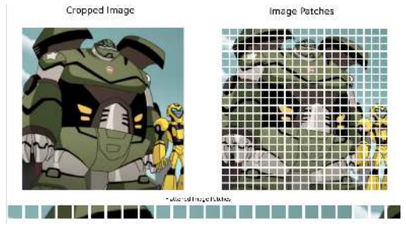


Figure 8: ViT splits an image into fixed-size patches, linearly embeds each of them, adds position embeddings, and feeds the resulting sequence of vectors to a standard Transformer encoder.

Spatial-patch Compression. This involves transforming video frames into fixed-size patches, akin to the methodologies employed in ViT [15] and MAE [33] (see Figure 8), before encoding them into a latent space. This approach is particularly effective for accommodating videos of varying resolutions and aspect ratios, as it encodes entire frames through the processing of individual patches. Subsequently, these spatial tokens are organized in a temporal sequence to create a spatial-temporal latent representation. This technique highlights several critical considerations:

用于在生成式人工智能中获得高质量结果的训练数据。 Sora 的训练方法与Richard Sutton提出的{T}HEBITTERLESSON原则相契合，该原则认为利用计算能力而非人类设计的特征能够打造出更高效且更具灵活性的人工智能系统。正如扩散变换器最初的设计理念注重简洁性与可扩展性一样，Sora通过使用原始尺寸的数据进行训练的策略，摒弃了传统人工智能对人类抽象概念的依赖，转而采用一种能随计算能力提升而扩展的通用方法。在本节的其余部分，我们将尝试逆向分析Sora的架构设计，并探讨实现这一卓越功能的相关技术。

3.2.2 统一的视觉表示方法

为有效处理包括图像以及时长、分辨率和宽高比各不相同的视频在内的多样化视觉输入，一种关键的方法是将各类视觉数据转换为统一的表示形式，这有助于生成模型的大规模训练。具体而言，Sora通过先将视频压缩到低维潜在空间，再将该表示分解为时空块的方式来对视频进行分块处理。然而，Sora的技术报告 [3] 仅阐述了高层次的概念，导致研究界难以对其进行复现。在本节中，我们将尝试逆向分析其可能的构成要素和技术路径。此外，我们还将结合现有文献中的见解，探讨能够实现与Sora类似功能的可行替代方案。

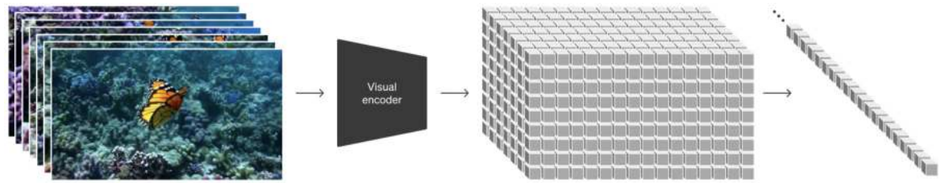


图7：从高层次来看，Sora首先将视频压缩为低维潜空间，再将该表示方法分解为时空块，从而将视频转换为块状结构。来源：Sora技术报告 [3]。

3.2.3 视频压缩网络

Sora所采用的视频压缩网络（或视觉编码器）旨在降低输入数据——尤其是原始视频的维度，进而输出在时间和空间上都经过压缩的潜在表示方法，如图7所示。根据技术报告中的参考资料，该压缩网络是基于VAE或向量量化VAE（VQ-VAE）构建的[32]。然而，如技术报告所述，若不采用尺寸调整和裁剪手段，VAE很难将任意大小的视觉数据映射到统一且固定大小的潜在空间中。为此，我们总结了两种不同的实现方案来解决这一问题：

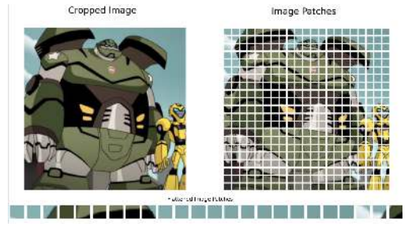


图8：ViT模型将图像分割成固定大小的块，对每个块进行线性嵌入，再加入位置嵌入，随后将生成的向量序列输入到标准的Transformer编码器中。

空间块压缩。该方法是将视频帧转换为固定大小的块，其思路与ViT [15] 和MAE [33] （见图8）中采用的方法类似，随后将这些块编码到潜在空间中。由于该方式能够通过整体编码整个帧，因此特别适合处理分辨率和宽高比各异的视频。

对各个图像块的单独处理。随后，这些空间标记会被按时间顺序排列，从而构建出时空类型的潜在表示。该技术凸显了几个关键的考量因素：

Temporal dimension variability – given the varying durations of training videos, the temporal dimension of the latent space representation cannot be fixed. To address this, one can either sample a specific number of frames (padding or temporal interpolation [34] may be needed for much shorter videos) or define a universally extended (super long) input length for subsequent processing (more details are described in Section 3.2.4); *Utilization of pre-trained visual encoders* – for processing videos of high resolution, leveraging existing pre-trained visual encoders, such as the VAE encoder from Stable Diffusion [19], is advisable for most researchers while Sora's team is expected to train their own compression network with a decoder (the video generator) from scratch via the manner employed in training latent diffusion models [19, 35, 36]. These encoders can efficiently compress large-size patches (e.g., 256×256), facilitating the management of large-scale data; *Temporal information aggregation* – since this method primarily focuses on spatial patch compression, it necessitates an additional mechanism for aggregating temporal information within the model. This aspect is crucial for capturing dynamic changes over time and is further elaborated in subsequent sections (see details in Section 3.3.1 and Figure 14).

Spatial-temporal-patch Compression. This technique is designed to encapsulate both spatial and temporal dimensions of video data, offering a comprehensive representation. This technique extends beyond merely analyzing static frames by considering the movement and changes across frames, thereby capturing the video's dynamic aspects. The utilization of 3D convolution emerges as a straightforward and potent method for achieving this integration [37]. The graphical illustration and the comparison against pure spatial-pachifying are depicted in Figure 9. Similar to spatial-patch compression, employing spatial-temporal-patch compression with predetermined convolution kernel parameters – such as fixed kernel sizes, strides, and output channels – results in variations in the dimensions of the latent space due to the differing characteristics of video inputs. This variability is primarily driven by the diverse durations and resolutions of the videos being processed. To mitigate this challenge, the approaches adopted for spatial patchification are equally applicable and effective in this context.

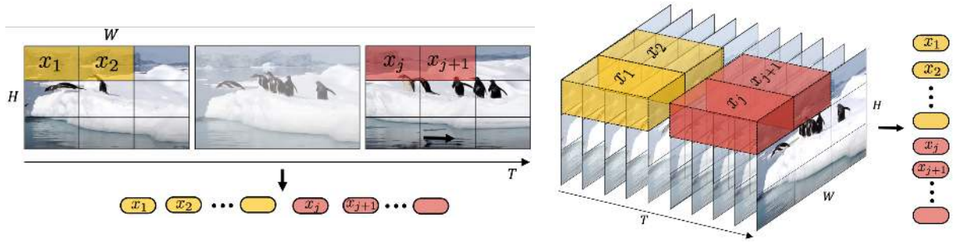


Figure 9: Comparison between different patchification for video compression. Source: ViViT [38]. (**Left**) Spatial patchification simply samples n_t frames and embeds each 2D frame independently following ViT. (**Right**) Spatial-temporal patchification extracts and linearly embeds non-overlapping or overlapping tubelets that span the spatiotemporal input volume.

In summary, we reverse engineer the two patch-level compression approaches based on VAE or its variant like VQ-VQE because operations on patches are more flexible to process different types of videos. Since Sora aims to generate high-fidelity videos, a large patch size or kernel size is used for efficient compression. Here, we expect that fixed-size patches are used for simplicity, scalability, and training stability. But varying-size patches could also be used [39] to make the dimension of the whole frames or videos in latent space consistent. However, it may result in invalid positional encoding, and cause challenges for the decoder to generate videos with varying-size latent patches.

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时空块压缩。该技术旨在同时封装视频数据的空间维度与时间维度，从而实现全面的表示方法。它不仅分析静态帧，还考虑帧与帧之间的运动变化，进而捕捉视频的动态特征。采用3D卷积是一种简单且高效的手段，可用于实现这种整合 [37]。相关的图形说明以及与纯空间块压缩方法的对比可见于图9中。与空间块压缩类似，当使用具有预定卷积核参数的时空块压缩方法——例如固定的卷积核大小、步长和输出通道数——时，由于视频输入的特性差异，潜在空间的维度也会出现变化。这种变化主要源于所处理视频的不同时长和分辨率。为应对这一挑战，用于空间块压缩的相关方法在此场景下同样适用且有效。

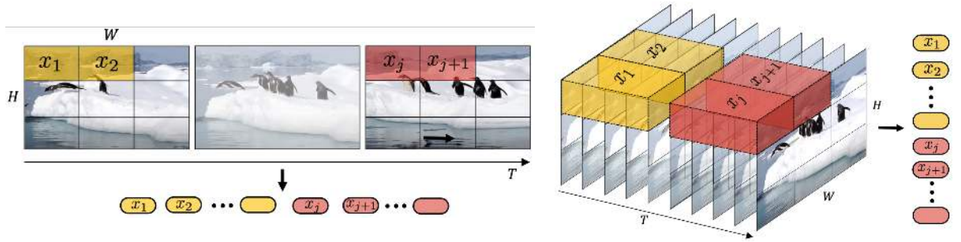


图9：不同视频压缩用分块处理方法的对比。来源：ViViT [38]。(**左侧**) 空间分块处理仅对 n_t 帧进行采样，然后按照ViT模型的方式独立嵌入每一帧2D图像。(**右侧**) 时空分块处理则提取跨越时空输入空间的非重叠或重叠的小管状结构，并对其进行线性嵌入。

总而言之，我们基于VAE或其变体如VQ-VQE反推了两种基于块级的压缩方法，因为对块进行操作在处理不同类型的视频时更具灵活性。由于Sora的目标是生成高保真度的视频，因此会采用较大的块尺寸或核尺寸以实现高效压缩。在此，我们认为使用固定大小的块是为了简化流程、提升可扩展性并保证训练稳定性。不过，使用不同大小的块 [39] 也能让潜在空间中整个帧或视频的维度保持一致。然而，这可能会导致位置编码失效，进而给解码器生成由不同大小潜在块构成的视频带来困难。

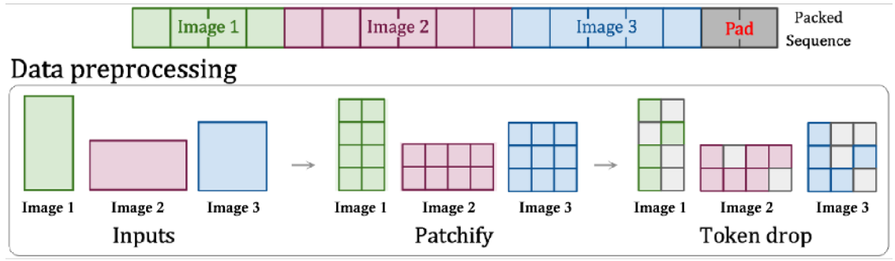


Figure 10: Patch packing enables variable resolution images or videos with preserved aspect ratio.6 Token dropping somehow could be treated as data augmentation. Source: NaViT [40].

3.2.4 Spacetime Latent Patches

There is a pivotal concern remaining in the compression network part: How to handle the variability in latent space dimensions (i.e., the number of latent feature chunks or patches from different video types) before feeding patches into the input layers of the diffusion transformer. Here, we discuss several solutions.

Based on Sora's technical report and the corresponding references, **patch n' pack (PNP)** [40] is likely the solution. PNP packs multiple patches from different images in a single sequence as shown in Figure 10. This method is inspired by example packing used in natural language processing [41] that accommodates efficient training on variable length inputs by dropping tokens. Here the patchification and token embedding steps need to be completed in the compression network, but Sora may further patchify the latent for transformer token as Diffusion Transformer does [4]. Regardless there is a second-round patchification or not, we need to address two concerns, how to pack those tokens in a compact manner and how to control which tokens should be dropped. For the first concern, a simple greedy approach is used which adds examples to the first sequence with enough remaining space. Once no more example can fit, sequences are filled with padding tokens, yielding the fixed sequence lengths needed for batched operations. Such a simple packing algorithm can lead to significant padding, depending on the distribution of the length of inputs. On the other hand, we can control the resolutions and frames we sample to ensure efficient packing by tuning the sequence length and limiting padding. For the second concern, an intuitive approach is to drop the similar tokens [42, 43, 33, 44] or, like PNP, apply dropping rate schedulers. However, it is worth noting that ***3D Consistency*** is one of the nice properties of Sora. Dropping tokens may ignore fine-grained details during training. Thus, we believe that OpenAI is likely to use a super long context window and pack all tokens from videos although doing so is computationally expensive e.g., the multi-head attention [45, 46] operator exhibits quadratic cost in sequence length. Specifically, spacetime latent patches from a long-duration video can be packed in one sequence while the ones from several short-duration videos are concatenated in the other sequence.

3.2.5 Discussion

We discuss two technical solutions to data pre-processing that Sora may use. Both solutions are performed at the patch level due to the characteristics of flexibility and scalability for modeling. Different from previous approaches where videos are resized, cropped, or trimmed to a standard size, Sora trains on data at its native size. Although there are several benefits (see detailed analysis in Section 3.2.1), it brings some technical challenges, among which one of the most significant is that neural networks cannot inherently process visual data of variable durations, resolutions, and aspect ratios. Through reverse engineering, we believe that Sora firstly compresses visual patches into low-dimensional latent representations, and arranges such latent patches or further patchified latent patches in a sequence, then injects noise into these latent patches

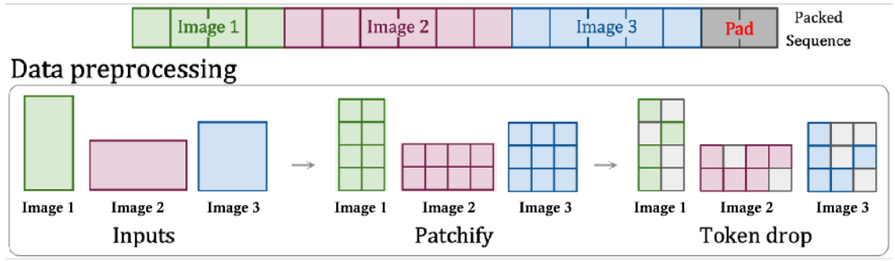


图10：通过拼块打包技术，可以生成保持宽高比的可变分辨率图像或视频。6 某种程度的令牌丢弃可被视为数据增强手段。来源：NaViT [40]。

3.2.4 时空潜在补丁

在压缩网络部分仍存在一个关键问题：在将各个特征块输入到扩散变换器的输入层之前，如何处理潜在空间维度上的差异（即不同视频类型所对应的潜在特征块或补丁的数量）。在此，我们探讨了几种解决方案。

Based on Sora's technical report and the corresponding references, **patch n' pack (PNP)** [40] is likely the solution. PNP packs multiple patches from different images in a single sequence as shown in Figure 10. This method is inspired by example packing used in natural language processing [41] that accommodates efficient training on variable length inputs by dropping tokens. Here the patchification and token embedding steps need to be completed in the compression network, but Sora may further patchify the latent for transformer token as Diffusion Transformer does [4]. Regardless there is a second-round patchification or not, we need to address two concerns, how to pack those tokens in a compact manner and how to control which tokens should be dropped. For the first concern, a simple greedy approach is used which adds examples to the first sequence with enough remaining space. Once no more example can fit, sequences are filled with padding tokens, yielding the fixed sequence lengths needed for batched operations. Such a simple packing algorithm can lead to significant padding, depending on the distribution of the length of inputs. On the other hand, we can control the resolutions and frames we sample to ensure efficient packing by tuning the sequence length and limiting padding. For the second concern, an intuitive approach is to drop the similar tokens [42, 43, 33, 44] or, like PNP, apply dropping rate schedulers. However, it is worth noting that ***3D Consistency*** is one of the nice properties of Sora. Dropping tokens may ignore fine-grained details during training. Thus, we believe that OpenAI is likely to use a super long context window and pack all tokens from videos although doing so is computationally expensive e.g., the multi-head attention [45, 46] operator exhibits quadratic cost in sequence length. Specifically, spacetime latent patches from a long-duration video can be packed in one sequence while the ones from several short-duration videos are concatenated in the other sequence.

3.2.5 讨论

We discuss two technical solutions to data pre-processing that Sora may use. Both solutions are performed at the patch level due to the characteristics of flexibility and scalability for modeling. Different from previous approaches where videos are resized, cropped, or trimmed to a standard size, Sora trains on data at its native size. Although there are several benefits (see detailed analysis in Section 3.2.1), it brings some technical challenges, among which one of the most significant is that neural networks cannot inherently process visual data of variable durations, resolutions, and aspect ratios. Through reverse engineering, we believe that Sora firstly compresses visual patches into low-dimensional latent representations, and arranges such latent patches or further patchified latent patches in a sequence, then injects noise into these latent patches

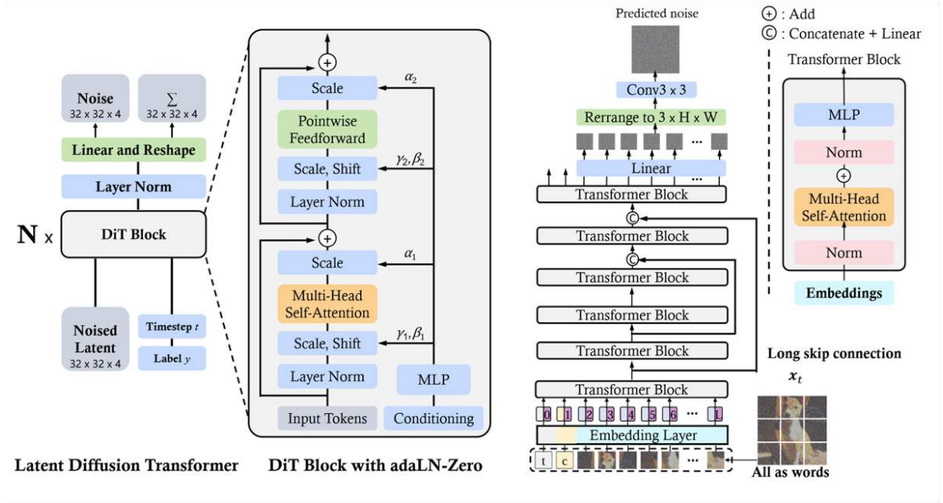


Figure 11: The overall framework of DiT (left) and U-ViT (right)

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before feeding them to the input layer of diffusion transformer. Spatial-temporal patchification is adopted by Sora because it is simple to implement, and it can effectively reduce the context length with high-information-density tokens and decrease the complexity of subsequent modeling of temporal information.

To the research community, we recommend using cost-efficient alternative solutions for video compression and representation, including utilizing pre-trained checkpoints (e.g., compression network) [47], shortening the context window, using light-weight modeling mechanisms such as (grouped) multi-query attention [48, 49] or efficient architectures (e.g. Mamba [50]), downsampling data and dropping tokens if necessary. The trade-off between effectiveness and efficiency for video modeling is an important research topic to be explored.

3.3 Modeling

3.3.1 Diffusion Transformer

Image Diffusion Transformer. Traditional diffusion models [51, 52, 53] mainly leverage convolutional U-Nets that include downsampling and upsampling blocks for the denoising network backbone. However, recent studies show that the U-Net architecture is not crucial to the good performance of the diffusion model. By incorporating a more flexible transformer architecture, the transformer-based diffusion models can use more training data and larger model parameters. Along this line, DiT [4] and U-ViT [54] are among the first works to employ vision transformers for latent diffusion models. As in ViT, DiT employs a multi-head self-attention layer and a pointwise feed-forward network interlaced with some layer norm and scaling layers. Moreover, as shown in Figure 11, DiT incorporates conditioning via adaptive layer norm (AdaLN) with an additional MLP layer for zero-initializing, which initializes each residual block as an identity function and thus greatly stabilizes the training process. The scalability and flexibility of DiT is empirically validated. DiT becomes the new backbone for diffusion models. In U-ViT, as shown in Figure 11, they treat all inputs,

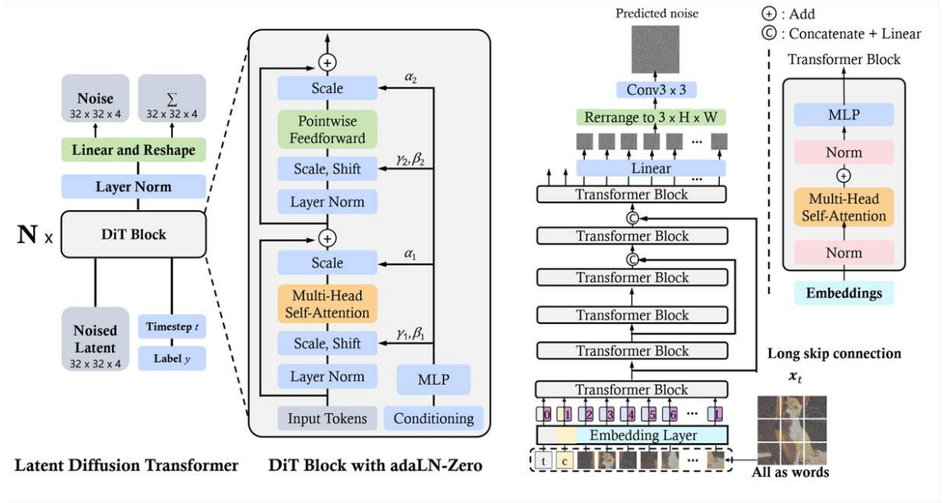


图11：DiT模型（左）与U-ViT模型（右）的总体架构

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在将它们输入到扩散变换器的输入层之前，会先进行预处理。Sora采用了时空分块处理方式，因为它实现起来较为简单，同时还能通过高信息密度标记有效缩短上下文长度，进而降低后续处理时间信息的模型复杂度。

对于研究界而言，我们建议采用成本更低的替代方案来实现视频的压缩与表示方法，例如利用预训练检查点（如压缩网络）。 [47],通过缩小上下文窗口，运用诸如（分组式）多查询注意力机制之类的轻量级建模机制 [48, 49] 或高效架构（例如Mamba模型 [50]），并在必要时对数据进行下采样并删除部分标记。在视频建模中如何平衡效果与效率，是值得深入探索的重要研究课题。

3.3 模型构建

3.3.1 扩散变换器

Image Diffusion Transformer. Traditional diffusion models [51, 52, 53] mainly leverage convolutional U-Nets that include downsampling and upsampling blocks for the denoising network backbone. However, recent studies show that the U-Net architecture is not crucial to the good performance of the diffusion model. By incorporating a more flexible transformer architecture, the transformer-based diffusion models can use more training data and larger model parameters. Along this line, DiT [4] and U-ViT [54] are among the first works to employ vision transformers for latent diffusion models. As in ViT, DiT employs a multi-head self-attention layer and a pointwise feed-forward network interlaced with some layer norm and scaling layers. Moreover, as shown in Figure 11, DiT incorporates conditioning via adaptive layer norm (AdaLN) with an additional MLP layer for zero-initializing, which initializes each residual block as an identity function and thus greatly stabilizes the training process. The scalability and flexibility of DiT is empirically validated. DiT becomes the new backbone for diffusion models. In U-ViT, as shown in Figure 11, they treat all inputs,

including time, condition, and noisy image patches, as tokens and propose long skip connections between the shallow and deep transformer layers. The results suggest that the downsampling and upsampling operators in CNN-based U-Net are not always necessary, and U-ViT achieves record-breaking FID scores in image and text-to-image generation.

Like Masked AutoEncoder (MAE) [33], Masked Diffusion Transformer (MDT) [55] incorporates mask latent modeling into the diffusion process to explicitly enhance contextual relation learning among object semantic parts in image synthesis. Specifically, as shown in Figure 12, MDT uses a side-interpolated for an additional masked token prediction task during training to boost the training efficiency and learn powerful context-aware positional embedding for inference. Compared to DiT [4], MDT achieves better performance and faster learning speed. In MDTv2 [56], they further improve MDT with a more efficient macro network with some u-shape long-shortcuts in the encoder and dense shortcuts from encoder input in the decoder. Furthermore, they also leverage a bag of training strategies like the Adan optimizer [57], Min-SNR weighting [58], dynamic masking ratio, and an improved power-cosine weighting for classifier-free guidance [59]. MDTv2 achieves a significant boost in synthesis performance and learning speed. Instead of using AdaLN (i.e., shifting and scaling) for time-conditioning modeling, Hatamizadeh et al. [60] introduce Diffusion Vision Transformers (DiffiT), which uses a time-dependent self-attention (TMSA) module to model dynamic denoising behavior over sampling time steps. Besides, DiffiT uses two hybrid hierarchical architectures for efficient denoising in the pixel space and the latent space, respectively, and achieves new state-of-the-art results across various generation tasks. Overall, these studies show promising results in employing vision transformers for image latent diffusion, paving the way for future studies for other modalities.

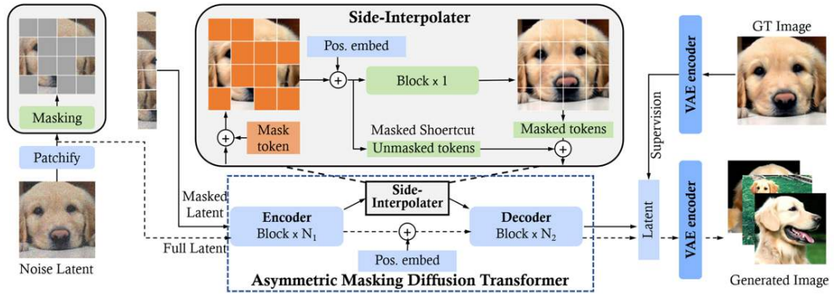


Figure 12: The overall framework of Masked Diffusion Transformer (MDT). A solid/dotted line indicates the training/inference process for each time step. Masking and side-interpolater are only used during training and are removed during inference.

Video Diffusion Transformer. Building upon the foundational works in text-to-image (T2I) diffusion models, recent research has been focused on realizing the potential of diffusion transformers for text-to-video (T2V) generation tasks. Due to the temporal nature of videos, key challenges for applying DiTs in the video domain are: *i) how to compress the video spatially and temporally to a latent space for efficient denoising*; *ii) how to convert the compressed latent to patches and feed them to the transformer;* and *iii) how to handle long-range temporal and spatial dependencies and ensure content consistency*. Please refer to Section 3.2.3 for the first challenge. In this section, we focus our discussion on transformer-based denoising network architectures designed to operate in the spatially and temporally compressed latent space. We give a detailed review of the two important works (Imagen Video [29] and Video LDM [36]) shown in the OpenAI `Sora` technique report and also briefly introduce some of recent promising approaches.

将这些时间、条件以及含噪声的图像块均作为标记，同时在浅层与深层Transformer层之间构建长跨度跳接结构。实验结果表明，基于CNN的U-Net中所使用的下采样与上采样操作并非始终必要，而U-ViT模型在图像生成及文本到图像生成任务中取得了打破纪录的FID分数。

Like Masked AutoEncoder (MAE) [33], Masked Diffusion Transformer (MDT) [55] incorporates mask latent modeling into the diffusion process to explicitly enhance contextual relation learning among object semantic parts in image synthesis. Specifically, as shown in Figure 12, MDT uses a side-interpolated for an additional masked token prediction task during training to boost the training efficiency and learn powerful context-aware positional embedding for inference. Compared to DiT [4], MDT achieves better performance and faster learning speed. In MDTv2 [56], they further improve MDT with a more efficient macro network with some u-shape long-shortcuts in the encoder and dense shortcuts from encoder input in the decoder. Furthermore, they also leverage a bag of training strategies like the Adan optimizer [57], Min-SNR weighting [58], dynamic masking ratio, and an improved power-cosine weighting for classifier-free guidance [59]. MDTv2 achieves a significant boost in synthesis performance and learning speed. Instead of using AdaLN (i.e., shifting and scaling) for time-conditioning modeling, Hatamizadeh et al. [60] introduce Diffusion Vision Transformers (DiffiT), which uses a time-dependent self-attention (TMSA) module to model dynamic denoising behavior over sampling time steps. Besides, DiffiT uses two hybrid hierarchical architectures for efficient denoising in the pixel space and the latent space, respectively, and achieves new state-of-the-art results across various generation tasks. Overall, these studies show promising results in employing vision transformers for image latent diffusion, paving the way for future studies for other modalities.

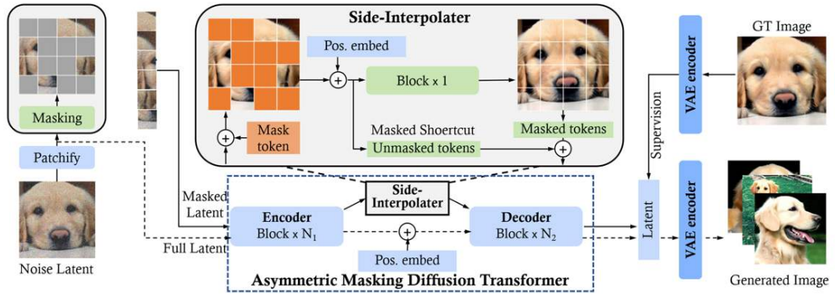
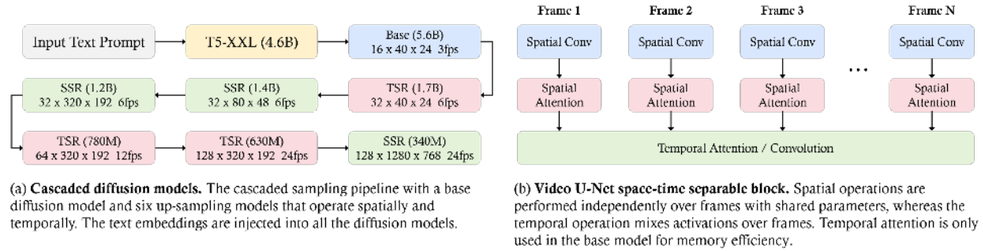


图12：掩码扩散变换器（MDT）的整体架构。实线/虚线分别表示每个时间步的训练过程与推理过程。掩码处理和侧向插值器仅在训练阶段使用，在推理阶段则会被移除。

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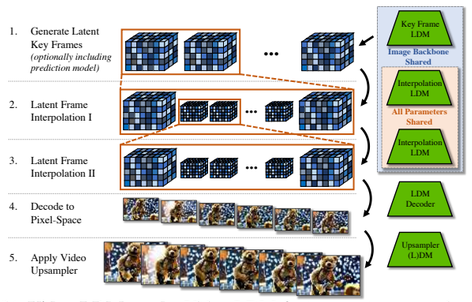


(a) **Cascaded diffusion models.** The cascaded sampling pipeline with a base diffusion model and six up-sampling models that operate spatially and temporally. The text embeddings are injected into all the diffusion models.

(b) **Video U-Net space-time separable block.** Spatial operations are performed independently over frames with shared parameters, whereas the temporal operation mixes activations over frames. Temporal attention is only used in the base model for memory efficiency.

Figure 13: The overall framework of Imagen Video. Source: Imagen Video [29].

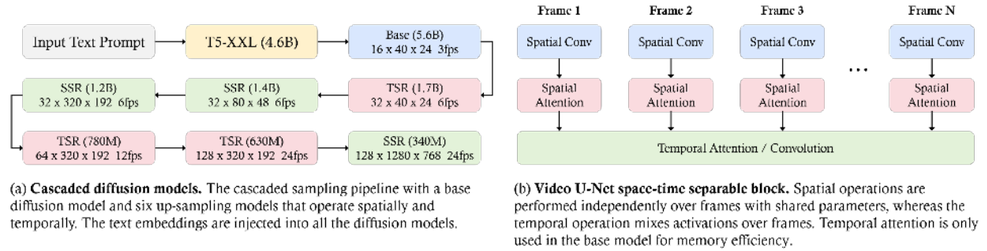
Imagen Video [29], a text-to-video generation system developed by Google Research, utilizes a cascade of diffusion models, which consists of 7 sub-models that perform text-conditional video generation, spatial super-resolution, and temporal super-resolution, to transform textual prompts into high-definition videos. As shown in Figure 13, firstly, a frozen T5 text encoder generates contextual embeddings from the input text prompt. These embeddings are critical for aligning the generated video with the text prompt and are injected into all models in the cascade, in addition to the base model. Subsequently, the embedding is fed to the base model for low-resolution video generation, which is then refined by cascaded diffusion models to increase the resolution. The base video and super-resolution models use a 3D U-Net architecture in a spatial-temporal separable fashion. This architecture weaves temporal attention and convolution layers with spatial counterparts to efficiently capture inter-frame dependencies. It employs v-prediction parameterization for numerical stability and conditioning augmentation to facilitate parallel training across models. The process involves joint training on both images and videos, treating each image as a frame to leverage larger datasets, and using classifier-free guidance [59] to enhance prompt fidelity. Progressive distillation [61] is applied to streamline the sampling process, significantly reducing the computational load while maintaining perceptual quality. Combining these methods and techniques allows Imagen Video to generate videos with not only high fidelity but also remarkable controllability, as demonstrated by its ability to produce diverse videos, text animations, and content in various artistic styles.



(a) **Additional temporal layer**. A pre-trained LDM is turned into a video generator by inserting temporal layers that learn to align frames into temporally consistent sequences. During optimization, the image backbone θ remains fixed and only the parameters ϕ of the temporal layers l_ϕ^i are trained.

(b) **Video LDM stack.** Video LDM first generates sparse key frames and then temporally interpolates twice with the same latent diffusion models to achieve a high frame rate. Finally, the latent video is decoded to pixel space, and optionally, a video upsampler diffusion model is applied.

Figure 14: The overall framework of Video LDM. Source: Video LDM [36].

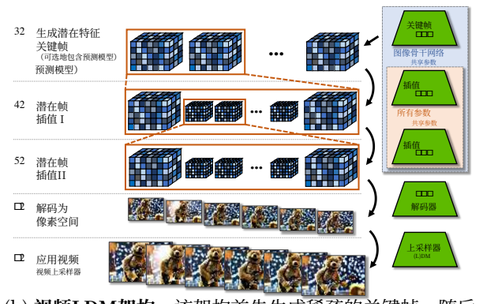


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图13：Imagen Video模型的整体架构。来源：Imagen Video [29]。

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(a) **额外的时间层**。通过插入能够学习将帧序排列为时间上连贯序列的时间层，预训练好的LDM算法便被转化为视频生成器。在优化过程中，图像主干网络 θ 保持不变，仅有时间层的参数 ϕ 会进行训练l_ϕ^i。

(b) **视频LDM架构**。该架构首先生成稀疏的关键帧，随后使用相同的潜在扩散模型进行两次时间插值，从而实现较高的帧率。最后，将潜在视频解码为像素空间，必要时还可应用视频上采样器扩散模型。

图14：视频LDM的整体框架。来源：Video LDM [36]。

Blattmann et al. [36] propose to turn a 2D Latent Diffusion Model into a Video Latent Diffusion Model (Video LDM). They achieve this by adding some post-hoc temporal layers among the existing spatial layers into both the U-Net backbone and the VAE decoder that learns to align individual frames. These temporal layers are trained on encoded video data, while the spatial layers remain fixed, allowing the model to leverage large image datasets for pre-training. The LDM's decoder is fine-tuned for temporal consistency in pixel space and temporally aligning diffusion model upsamplers for enhanced spatial resolution. To generate very long videos, models are trained to predict a future frame given a number of context frames, allowing for classifier-free guidance during sampling. To achieve high temporal resolution, the video synthesis process is divided into key frame generation and interpolation between these keyframes. Following cascaded LDMs, a DM is used to further scale up the Video LDM outputs by four times, ensuring high spatial resolution while maintaining temporal consistency. This approach enables the generation of globally coherent long videos in a computationally efficient manner. Additionally, the authors demonstrate the ability to transform pre-trained image LDMs (e.g., Stable Diffusion) into text-to-video models by training only the temporal alignment layers, achieving video synthesis with resolutions up to 1280×2048px.

Recently, there are also some other promising works that show superior performance in high-quality and temporal-consistent video generation. Similar to Imagen Video, LAVIE [62] proposes to transform a T2I model into a T2V model with cascaded LDMs, with two key insights: i) using simply temporal self-attention and RoPE [63] can adequately capture temporal correlations inherent in video data; ii) joint image-video fine-tuning enable more stable training and avoid catastrophic forgetting. GenTron [64] adapt pre-trained class-conditioning DiTs to textual conditioning and explore different conditioning mechanisms, where they find that cross attention is a better conditioning method compared to previous adaLM-Zero for free-form textual conditioning. Furthermore, by designing motion-free masking in the inserted temporal self-attention layers, GenTron is able to do efficient image-video joint training with control on the temporal modeling ability in the video diffusion process. Window Attention Latent Transformer (W.A.L.T) [65] proposes to unify the latent space of both videos and images with 3D causal CNN and a special coding strategy for the first frame. Moreover, W.A.L.T uses non-overlapping, window-restricted spatial and spatiotemporal attention to lower the computational cost and provide flexibility for image-video joint training. Latent Diffusion Transformer (Latte) [66] conducts extensive experiments among four model variants from the perspective of decomposing spatial and temporal information in video data. Furthermore, Latte studies the effect of a wide range of training strategies and modules in a DiT-based video generation system, including conditioning mechanisms and temporal positional embedding. Instead of using separate compression for spatial and temporal dimensions in previous works, SnapVideo [67] proposes to extend EDM framework [68] for high-dimensional video data. Furthermore, SnapVideo treats images as T frames videos with infinite frame rates and introduces a variable frame-rate training procedure to bridge the modality gap. For scaling up to billion-parameters size, SnapVideo uses an efficient transformer-based architecture, FITs [69], as a scalable video generator, and a model cascade consisting of a first-stage lower-solution base model (36×64px) and a high-solution upsampler (288×512px) with corrupt training strategy.

3.3.2 Discussion

One end-to-end model or cascade structure. `Sora` can generate high-resolution videos. By reviewing existing works and our reverse engineering, we speculate that `Sora` also leverages cascade diffusion model architecture [70], which is composed of a base model and many spatial-temporal refiner models. The attention modules are unlikely to be heavily used in the based diffusion model and low-resolution diffusion model, considering the high computation cost and limited performance gain of using attention machines in high-resolution cases. For spatial and temporal scene consistency, as previous works show that temporal consistency is more important than spatial consistency for video/scene generation, `Sora` is likely to leverage an efficient training strategy by using longer video (for temporal consistency) with lower resolution. More-

Blattmann et al. [36] propose to turn a 2D Latent Diffusion Model into a Video Latent Diffusion Model (Video LDM). They achieve this by adding some post-hoc temporal layers among the existing spatial layers into both the U-Net backbone and the VAE decoder that learns to align individual frames. These temporal layers are trained on encoded video data, while the spatial layers remain fixed, allowing the model to leverage large image datasets for pre-training. The LDM's decoder is fine-tuned for temporal consistency in pixel space and temporally aligning diffusion model upsamplers for enhanced spatial resolution. To generate very long videos, models are trained to predict a future frame given a number of context frames, allowing for classifier-free guidance during sampling. To achieve high temporal resolution, the video synthesis process is divided into key frame generation and interpolation between these keyframes. Following cascaded LDMs, a DM is used to further scale up the Video LDM outputs by four times, ensuring high spatial resolution while maintaining temporal consistency. This approach enables the generation of globally coherent long videos in a computationally efficient manner. Additionally, the authors demonstrate the ability to transform pre-trained image LDMs (e.g., Stable Diffusion) into text-to-video models by training only the temporal alignment layers, achieving video synthesis with resolutions up to 1280×2048px.

Recently, there are also some other promising works that show superior performance in high-quality and temporal-consistent video generation. Similar to Imagen Video, LAVIE [62] proposes to transform a T2I model into a T2V model with cascaded LDMs, with two key insights: i) using simply temporal self-attention and RoPE [63] can adequately capture temporal correlations inherent in video data; ii) joint image-video fine-tuning enable more stable training and avoid catastrophic forgetting. GenTron [64] adapt pre-trained class-conditioning DiTs to textual conditioning and explore different conditioning mechanisms, where they find that cross attention is a better conditioning method compared to previous adaLM-Zero for free-form textual conditioning. Furthermore, by designing motion-free masking in the inserted temporal self-attention layers, GenTron is able to do efficient image-video joint training with control on the temporal modeling ability in the video diffusion process. Window Attention Latent Transformer (W.A.L.T) [65] proposes to unify the latent space of both videos and images with 3D causal CNN and a special coding strategy for the first frame. Moreover, W.A.L.T uses non-overlapping, window-restricted spatial and spatiotemporal attention to lower the computational cost and provide flexibility for image-video joint training. Latent Diffusion Transformer (Latte) [66] conducts extensive experiments among four model variants from the perspective of decomposing spatial and temporal information in video data. Furthermore, Latte studies the effect of a wide range of training strategies and modules in a DiT-based video generation system, including conditioning mechanisms and temporal positional embedding. Instead of using separate compression for spatial and temporal dimensions in previous works, SnapVideo [67] proposes to extend EDM framework [68] for high-dimensional video data. Furthermore, SnapVideo treats images as T frames videos with infinite frame rates and introduces a variable frame-rate training procedure to bridge the modality gap. For scaling up to billion-parameters size, SnapVideo uses an efficient transformer-based architecture, FITs [69], as a scalable video generator, and a model cascade consisting of a first-stage lower-solution base model (36×64px) and a high-solution upsampler (288×512px) with corrupt training strategy.

3.3.2 讨论

要么是端到端的单一体模型，要么是级联结构。Sora具备生成高分辨率视频的能力。通过梳理现有研究成果以及我们的逆向分析，我们推测Sora还采用了由基础模型与众多时空优化模型构成的级联扩散模型架构 [70], 。考虑到在处理高分辨率内容时使用注意力机制会带来高昂的计算成本且性能提升有限，这类注意力模块在基础扩散模型以及低分辨率扩散模型中很可能不会被大量运用。至于场景的时空一致性问题，正如先前的研究所示，在视频/场景生成过程中时间一致性比空间一致性更为重要，因此Sora很可能会采用一种高效的训练策略，即使用分辨率较低但长度更长的视频来保障时间一致性。更多——

over, Sora is likely to use a v-parameterization diffusion model [61], considering its superior performance among others that predict the original latent x or the noise ϵ.

On the compression of the latent encoder. For training efficiency, most of the existing works leverage the pre-trained VAE encoder of Stable Diffusions [71, 72], a pre-trained 2D diffusion model, as an initialized model checkpoint. However, the encoder lacks the ability to compress temporally. Even though some works propose to only fine-tune the decoder for handling temporal information, the decoder's performance of dealing with video temporal data in the compressed latent space remains sub-optimal. Based on the technique report, our reverse engineering shows that, instead of using an existing pre-trained Image VAE encoder with some post-hoc temporal tuning, it is likely that Sora uses a spatial-temporal VAE encoder, jointly trained on image and video data, to compress the spatial and temporal information altogether.

3.4 Language Instruction Following

Users primarily engage with generative AI models through natural language instructions, known as text prompts [73, 74]. Model instruction tuning aims to enhance AI models' capability to follow prompts accurately. This improved capability in prompt following enables models to generate output that more closely resembles human responses to natural language queries. We start our discussion with a review of instruction following techniques for large language models (LLMs) and text-to-image models such as DALL·E 3. To enhance the text-to-video model's ability to follow text instructions, Sora utilizes an approach similar to that of DALL·E 3. The approach involves training a descriptive captioner and utilizing the captioner's generated data for fine-tuning. As a result of instruction tuning, Sora is able to accommodate a wide range of user requests, ensuring meticulous attention to the details in the instructions and generating videos that precisely meet users' needs.

3.4.1 Large Language Models

The capability of LLMs to follow instructions has been extensively explored [75, 76, 77]. This ability allows LLMs to read, understand, and respond appropriately to instructions describing an unseen task without examples. Prompt following ability is obtained and enhanced by fine-tuning LLMs on a mixture of tasks formatted as instructions[75, 77], known as instruction tuning. Wei et al. [76] showed that instruction-tuned LLMs significantly outperform the untuned ones on unseen tasks. The instruction-following ability transforms LLMs into general-purpose task solvers, marking a paradigm shift in the history of AI development.

3.4.2 Text-to-Image

The instruction following in DALL·E 3 is addressed by a caption improvement method with a hypothesis that the quality of text-image pairs that the model is trained on determines the performance of the resultant text-to-image model [78]. The poor quality of data, particularly the prevalence of noisy data and short captions that omit a large amount of visual information, leads to many issues such as neglecting keywords and word order, and misunderstanding the user intentions [21]. The caption improvement approach addresses these issues by re-captioning existing images with detailed, descriptive captions. The approach first trains an image captioner, which is a vision-language model, to generate precise and descriptive image captions. The resulting descriptive image captions by the captioner are then used to fine-tune text-to-image models. Specifically, DALL·E 3 follows contrastive captioners (CoCa) [79] to jointly train an image captioner with a CLIP [26] architecture and a language model objective. This image captioner incorporates an image encoder a unimodal text encoder for extracting language information, and a multimodal text decoder. It first employs a contrastive loss between unimodal image and text embeddings, followed by a captioning loss for the multimodal decoder's outputs. The resulting image captioner is further fine-tuned on a highly detailed description of images covering main objects, surroundings, backgrounds, texts, styles, and colorations. With this step, the image captioner is able to generate detailed descriptive captions for the images. The training

之上，Sora 很可能会采用一种 v具有参数化结构的扩散模型 [61],，因为它在预测原始潜在特征 x 或噪声 ϵ的其他同类模型中表现更为优异。

关于潜在编码器的压缩问题。为提升训练效率，现有大多数研究都采用预训练好的Stable Diffusions [71, 72], 的VAE编码器——一种预训练的2D扩散模型——作为初始模型检查点。然而，这种编码器缺乏压缩时间信息的能力。尽管有一些研究提出仅对解码器进行微调以处理时间信息，但该解码器在压缩后的潜在空间中处理视频时间数据的性能依然不尽如人意。根据技术报告，我们的逆向工程分析表明，Sora很可能并未使用经过事后时间信息调整的现有预训练图像VAE编码器，而是采用了同时在图像和视频数据上共同训练的空间-时间VAE编码器，从而一次性压缩空间与时间信息。

3.4 语言指令遵循能力

用户主要通过自然语言指令——即文本提示词 [73, 74]——与生成型人工智能技术进行交互。模型指令微调旨在提升人工智能技术准确遵循提示词的能力。这种得到强化的指令遵循能力使得模型能够生成更接近人类对自然语言查询响应的内容。我们首先回顾一下大语言模型（LLM）以及DALL·E 3这类文本转图像模型的指令遵循技术，以此作为讨论的起点。为了提升文本转视频模型遵循文本指令的能力，Sora采用了与DALL·E 3类似的方法。该方法包括训练一个描述性字幕生成器，并利用该生成器产生的数据来进行进一步微调。通过指令微调，Sora能够应对各种不同的用户需求，仔细关注指令中的细节，从而生成完全符合用户要求的视频。

3.4.1 大语言模型

已有大量研究致力于探索大语言模型遵循指令的能力 [75, 76, 77]。这一能力使大语言模型能够阅读、理解那些未提供示例的指令，并据此给出恰当的回应。通过针对以指令形式呈现的各种任务对大语言模型进行微调，即所谓的指令微调，可以提升其遵循指令的能力[75, 77], 。Wei等人 [76] 的研究表明，在处理未见过的任务时，经过指令微调的大语言模型性能显著优于未经微调的模型。这种遵循指令的能力让大语言模型具备了处理通用任务的潜力，从而为人工智能技术的发展带来了范式上的转变。

3.4.2 文本生成图像

The instruction following in DALL·E 3 is addressed by a caption improvement method with a hypothesis that the quality of text-image pairs that the model is trained on determines the performance of the resultant text-to-image model [78]. The poor quality of data, particularly the prevalence of noisy data and short captions that omit a large amount of visual information, leads to many issues such as neglecting keywords and word order, and misunderstanding the user intentions [21]. The caption improvement approach addresses these issues by re-captioning existing images with detailed, descriptive captions. The approach first trains an image captioner, which is a vision-language model, to generate precise and descriptive image captions. The resulting descriptive image captions by the captioner are then used to fine-tune text-to-image models. Specifically, DALL·E 3 follows contrastive captioners (CoCa) [79] to jointly train an image captioner with a CLIP [26] architecture and a language model objective. This image captioner incorporates an image encoder a unimodal text encoder for extracting language information, and a multimodal text decoder. It first employs a contrastive loss between unimodal image and text embeddings, followed by a captioning loss for the multimodal decoder's outputs. The resulting image captioner is further fine-tuned on a highly detailed description of images covering main objects, surroundings, backgrounds, texts, styles, and colorations. With this step, the image captioner is able to generate detailed descriptive captions for the images. The training

dataset for the text-to-image model is a mixture of the re-captioned dataset generated by the image captioner and ground-truth human-written data to ensure that the model captures user inputs. This image caption improvement method introduces a potential issue: a mismatch between the actual user prompts and descriptive image descriptions from the training data. DALL·E 3 addresses this by *upsampling*, where LLMs are used to re-write short user prompts into detailed and lengthy instructions. This ensures that the model's text inputs received in inference time are consistent with those in model training.

3.4.3 Text-to-Video

To enhance the ability of instruction following, `Sora` adopts a similar caption improvement approach. This method is achieved by first training a video captioner capable of producing detailed descriptions for videos. Then, this video captioner is applied to all videos in the training data to generate high-quality (video, descriptive caption) pairs, which are used to fine-tune `Sora` to improve its instruction following ability.

`Sora`'s technical report [3] does not reveal the details about how the video captioner is trained. Given that the video captioner is a video-to-text model, there are many approaches to building it. A straightforward approach is to utilize CoCa architecture for video captioning by taking multiple frames of a video and feeding each frame into the image encoder [79], known as VideoCoCa [80]. VideoCoCa builds upon CoCa and re-uses the image encoder pre-trained weights and applies it independently on sampled video frames. The resulting frame token embeddings are flattened and concatenated into a long sequence of video representations. These flattened frame tokens are then processed by a generative pooler and a contrastive pooler, which are jointly trained with the contrastive loss and captioning loss. Other alternatives to building video captioners include mPLUG-2 [81], GIT [82], FrozenBiLM [83], and more. Finally, to ensure that user prompts align with the format of those descriptive captions in training data, `Sora` performs an additional prompt extension step, where GPT-4V is used to expand user inputs to detailed descriptive prompts.

3.4.4 Discussion

The instruction-following ability is critical for `Sora` to generate one-minute-long videos with intricate scenes that are faithful to user intents. According to `Sora`'s technical report [3], this ability is obtained by developing a captioner that can generate long and detailed captions, which are then used to train the model. However, the process of collecting data for training such a captioner is unknown and likely labor-intensive, as it may require detailed descriptions of videos. Moreover, the descriptive video captioner might hallucinate important details of the videos. We believe that how to improve the video captioner warrants further investigation and is critical to enhance the instruction-following ability of text-to-image models.

3.5 Prompt Engineering

Prompt engineering refers to the process of designing and refining the input given to an AI system, particularly in the context of generative models, to achieve specific or optimized outputs [84, 85, 86]. The art and science of prompt engineering involve crafting these inputs in a way that guides the model to produce the most accurate, relevant, and coherent responses possible.

3.5.1 Text Prompt

Text prompt engineering is vital in directing text-to-video models (e.g., `Sora` [3]) to produce videos that are visually striking while precisely meeting user specifications. This involves crafting detailed descriptions to instruct the model to effectively bridge the gap between human creativity and AI's execution capabilities [87]. The prompts for `Sora` cover a wide range of scenarios. Recent works (e.g., VoP [88],

文本到图像模型所使用的训练数据集是由图像描述生成器生成的重新标注数据集与真实的人类手写数据混合而成的，这样做的目的是确保模型能够准确捕捉用户的输入内容。这种图像描述优化方法却引出了一个潜在问题：实际的用户提示与训练数据中的描述性图像文字之间存在不一致。DALL·E 3通过上采样技术来解决这一问题，即利用大语言模型将简短的用户提示改写为详细且篇幅较长的指令，从而保证模型在推理阶段接收到的文本输入与训练时的输入保持一致。

3.4.3 文本转视频

为提升指令遵循能力，Sora采用了类似的字幕优化方法。该方法的实现步骤为：首先训练一个能够为视频生成详细描述的视频字幕生成器；随后将这个视频字幕生成器应用于训练数据中的所有视频，以此生成高质量的视频与描述性字幕对，进而用于微调Sora，从而提高其指令遵循能力。

`Sora`'s technical report [3] does not reveal the details about how the video captioner is trained. Given that the video captioner is a video-to-text model, there are many approaches to building it. A straightforward approach is to utilize CoCa architecture for video captioning by taking multiple frames of a video and feeding each frame into the image encoder [79], known as VideoCoCa [80]. VideoCoCa builds upon CoCa and re-uses the image encoder pre-trained weights and applies it independently on sampled video frames. The resulting frame token embeddings are flattened and concatenated into a long sequence of video representations. These flattened frame tokens are then processed by a generative pooler and a contrastive pooler, which are jointly trained with the contrastive loss and captioning loss. Other alternatives to building video captioners include mPLUG-2 [81], GIT [82], FrozenBiLM [83], and more. Finally, to ensure that user prompts align with the format of those descriptive captions in training data, `Sora` performs an additional prompt extension step, where GPT-4V is used to expand user inputs to detailed descriptive prompts.

3.4.4 讨论

对于Sora而言，具备出色的指令遵循能力至关重要，这让它能够生成时长为一分钟、场景复杂且完全符合用户意图的视频。根据Sora的技术报告 [3], , 这一能力是通过开发一种能够生成冗长详尽字幕的字幕生成器来实现的，这些字幕随后会被用于模型训练。然而，为训练此类字幕生成器而收集数据的具体流程尚不明确，而且很可能需要耗费大量人力，因为这可能需要针对视频提供极为详细的描述。此外，这种描述型视频字幕生成器还可能编造出视频中的重要细节。我们认为，如何改进视频字幕生成器值得进一步研究，这也是提升文本到图像模型指令遵循能力的关键所在。

3.5 提示工程

提示工程指的是设计并优化输入给人工智能系统的内容的流程，尤其是在生成式模型的背景下，以此获得特定或最优化的输出 [84, 85, 86]。提示工程的艺术与科学在于以恰当的方式构造这些输入，从而引导模型产生最准确、最相关且最连贯的响应。

3.5.1 文本提示词

在引导文本转视频模型（例如Sora [3]）生成既具有视觉冲击力又能精准符合用户要求的视频方面，提示工程发挥着至关重要的作用。它需要通过编写详尽的描述来指导模型，从而有效弥合人类创造力与人工智能执行能力之间的差距 [87]。针对Sora的文本提示词涵盖了极为多样的应用场景，近期也有相关研究成果涌现（例如VoP [88],）

Make-A-Video [28], and Tune-A-Video [89]) have shown how prompt engineering leverages model's natural language understanding ability to decode complex instructions and render them into cohesive, lively, and high-quality video narratives. As shown in Figure 15, "a stylish woman walking down a neon-lit Tokyo street..." is such a meticulously crafted text prompt that it ensures Sora to generate a video that aligns well with the expected vision. The quality of prompt engineering depends on the careful selection of words, the specificity of the details provided, and comprehension of their impact on the model's output. For example, the prompt in Figure 15 specifies in detail the actions, settings, character appearances, and even the desired mood and atmosphere of the scene.



Figure 15: A case study on prompt engineering for text-to-video generation, employing color coding to delineate the creative process. The text highlighted in blue describes the elements generated by Sora, such as the depiction of a stylish woman. In contrast, the text in yellow accentuates the model's interpretation of actions, settings, and character appearances, demonstrating how a meticulously crafted prompt is transformed into a vivid and dynamic video narrative.

3.5.2 Image Prompt

An image prompt serves as a visual anchor for the to-be-generated video's content and other elements such as characters, setting, and mood [90]. In addition, a text prompt can instruct the model to animate these elements by e.g., adding layers of movement, interaction, and narrative progression that bring the static image to life [27, 91, 92]. The use of image prompts allows Sora to convert static images into dynamic, narrative-driven videos by leveraging both visual and textual information. In Figure 16, we show AI-generated videos of "a Shiba Inu wearing a beret and turtleneck", "a unique monster family", "a cloud forming the word 'SORA"' and "surfers navigating a tidal wave inside a historic hall". These examples demonstrate what can be achieved by prompting Sora with DALL·E-generated images.

3.5.3 Video Prompt

Video prompts can also be used for video generation as demonstrated in [93, 94]. Recent works (e.g., Moonshot [95] and Fast-Vid2Vid [96]) show that good video prompts need to be specific and flexible. This ensures that the model receives clear direction on specific objectives, like the portrayal of particular objects and visual themes, and also allows for imaginative variations in the final output. For example, in the video extension tasks, a prompt could specify the direction (forward or backward in time) and the context or theme of the extension. In Figure 17(a), the video prompt instructs Sora to extend a video backward in time to explore the events leading up to the original starting point. When performing video-to-video editing through video prompts, as shown in Figure 17(b), the model needs to clearly understand the desired transformation, such as changing the video's style, setting or atmosphere, or altering subtle aspects like lighting or mood. In Figure 17(c), the prompt instructs Sora to connect videos while ensuring smooth transitions between objects in different scenes across videos.

Make-A-Video [28], and Tune-A-Video [89]) have shown how prompt engineering leverages model's natural language understanding ability to decode complex instructions and render them into cohesive, lively, and high-quality video narratives. As shown in Figure 15, "a stylish woman walking down a neon-lit Tokyo street..." is such a meticulously crafted text prompt that it ensures Sora to generate a video that aligns well with the expected vision. The quality of prompt engineering depends on the careful selection of words, the specificity of the details provided, and comprehension of their impact on the model's output. For example, the prompt in Figure 15 specifies in detail the actions, settings, character appearances, and even the desired mood and atmosphere of the scene.



图15：一项关于文本转视频生成领域提示工程的案例研究，该研究运用颜色编码来呈现创意生成过程。蓝色高亮显示的文字描述了Sora生成的元素，比如对一位时尚女子的刻画；而黄色文字则突出了模型对动作、场景及角色外貌的解读，从而展示了如何通过精心设计的提示将内容转化为生动且富有动态感的视频叙事。

3.5.2 图像提示词

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3.5.3 视频提示词

Video prompts can also be used for video generation as demonstrated in [93, 94]. Recent works (e.g., Moonshot [95] and Fast-Vid2Vid [96]) show that good video prompts need to be specific and flexible. This ensures that the model receives clear direction on specific objectives, like the portrayal of particular objects and visual themes, and also allows for imaginative variations in the final output. For example, in the video extension tasks, a prompt could specify the direction (forward or backward in time) and the context or theme of the extension. In Figure 17(a), the video prompt instructs Sora to extend a video backward in time to explore the events leading up to the original starting point. When performing video-to-video editing through video prompts, as shown in Figure 17(b), the model needs to clearly understand the desired transformation, such as changing the video's style, setting or atmosphere, or altering subtle aspects like lighting or mood. In Figure 17(c), the prompt instructs Sora to connect videos while ensuring smooth transitions between objects in different scenes across videos.

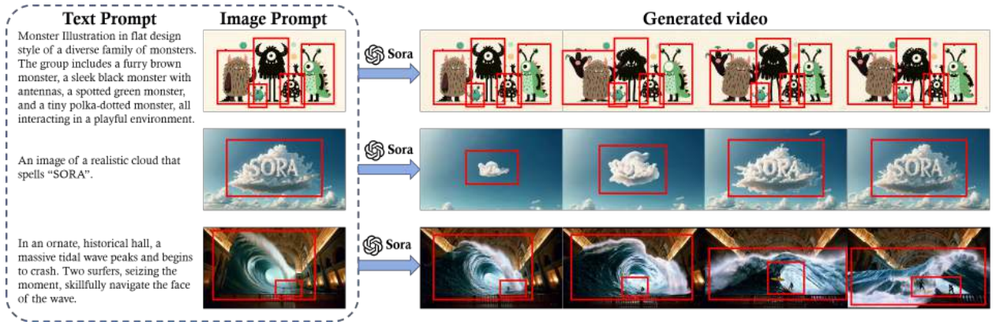


Figure 16: This example illustrates the image prompts to guide Sora's text-to-video model to generation. The red boxes visually anchor the key elements of each scene—monsters of varied designs, a cloud formation spelling "SORA", and surfers in an ornate hall facing a massive tidal wave.

3.5.4 Discussion

Prompt engineering allows users to guide AI models to generate content that aligns with their intent. As an example, the combined use of text, image, and video prompts enables Sora to create content that is not only visually compelling but also aligned well with users' expectations and intent. While previous studies on prompt engineering have been focused on text and image prompts for LLMs and LVMs [97, 98, 99], we expect that there will be a growing interest in video prompts for video generation models.

3.6 Trustworthiness

With the rapid advancement of sophisticated models such as ChatGPT [100], GPT4-V [101], and Sora [3], the capabilities of these models have seen remarkable enhancements. These developments have made significant contributions to improving work efficiency and propelling technological progress. However, these advancements also raise concerns about the potential for misuse of these technologies, including the generation of fake news [102, 103], privacy breaches [104], and ethical dilemmas [105, 106]. Consequently, the issue of trustworthiness in large models has garnered extensive attention from both the academic and industrial spheres, emerging as a focal point of contemporary research discussions.

3.6.1 Safety Concern

One primary area of focus is the model's safety, specifically its resilience against misuse and so-called "jailbreak" attacks, where users attempt to exploit vulnerabilities to generate prohibited or harmful content [107, 108, 109, 110, 111, 112, 113, 114, 115, 116]. For instance, AutoDAN [114], a novel and interpretable adversarial attack method based on gradient techniques, is introduced to enable system bypass. In a recent study, researchers explore two reasons why LLMs struggle to resist jailbreak attacks: competing objectives and mismatched generalization [117]. Besides textual attacks, visual jailbreak also threatens the safety of multimodal models (e.g., GPT-4V [101], and Sora [3]). A recent study [118] found that large multimodal models are more vulnerable since the continuous and high-dimensional nature of the additional visual input makes it weaker against adversarial attacks, representing an expanded attack surface.

3.6.2 Other Exploitation

Due to the large scale of the training dataset and training methodology of large foundation models (e.g., ChatGPT [100] and Sora [3]), the truthfulness of these models needs to be enhanced as the related issues like hallucination have been discussed widely [119]. Hallucination in this context refers to the models' tendency to generate responses that may appear convincing but are unfounded or false [107]. This phe-

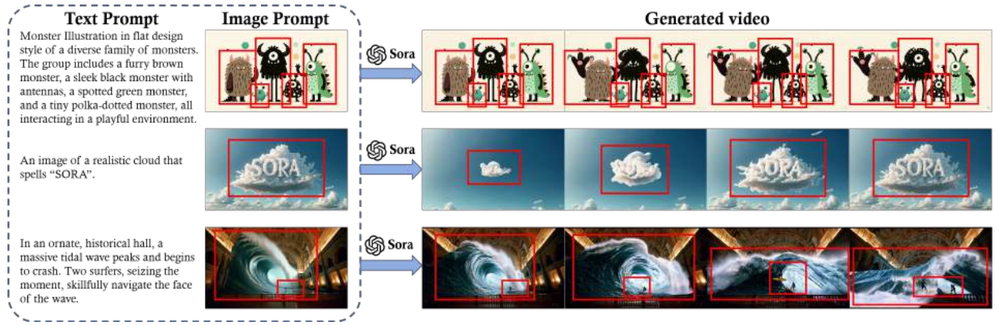


图16：该示例展示了用于引导Sora的文本转视频模型进行生成的图像提示词。红色框直观地标出了每个场景中的关键元素——造型各异的怪物、拼成“SORA”字样的云朵，以及身处华丽大厅中、正面对着巨大潮浪的冲浪者。

3.5.4 讨论

通过提示工程，用户能够引导人工智能技术生成符合其意图的内容。例如，结合使用文本、图像和视频提示词，就可以Sora 创造出不仅视觉效果出色，还能完美契合用户期望与意图的内容。尽管以往关于提示工程的研究多集中在面向大语言模型和多模态语言模型的文本及图像提示词上[97, 98, 99], 我们预计，针对视频生成模型的视频提示词将会越来越受到关注。

3.6 可信赖性

随着ChatGPT [100], 、GPT4-V [101], 以及Sora[3], 这类先进模型的飞速发展，它们的功能得到了极为显著的提升。这些进展为提高工作效率与推动技术进步做出了重要贡献。然而，这样的进步也引发了人们对这些技术可能被滥用的担忧，比如虚假新闻的生成 [102, 103], 、隐私泄露 [104], 以及伦理困境 [105, 106]等问题。正因如此，大型模型的可信度问题已经引起了学术界和工业界的广泛关注，成为当前研究讨论的焦点之一。

3.6.1 安全隐患

其中一个核心关注点在于该模型的安全性，尤其是其抵御滥用以及所谓“越狱”攻击的能力——在这种攻击中，用户会试图利用漏洞来生成被禁止或有害的内容[107, 108, 109, 110, 111, 112, 113, 114, 115, 116]。例如，AutoDAN [114], 便是一种基于梯度技术的新型且可解释的对抗性攻击方法，旨在帮助用户绕过系统防御。在最近的一项研究中，研究人员探讨了大语言模型难以抵御越狱攻击的两个原因：相互冲突的目标以及泛化能力的不足 [117]。除了文本层面的攻击之外，视觉越狱也会威胁到多模态模型（如GPT-4V [101], 和Sora [3]）的安全性。另一项最新研究 [118] 发现，由于额外的视觉输入具有连续性和高维度的特点，使得大型多模态模型在面对对抗性攻击时更为脆弱，进而也扩大了其受攻击的潜在面。

3.6.2 其他应用场景

由于大型基础模型（如ChatGPT [100] 和Sora [3]）的训练数据集规模庞大且训练方法复杂，加之诸如幻觉现象等相关问题已得到广泛探讨 [119], 这些模型的真实性亟需提升。在此语境下，幻觉现象指的是模型倾向于生成看似有理有据但实际上毫无依据或完全错误的回应 [107]。这一现象——

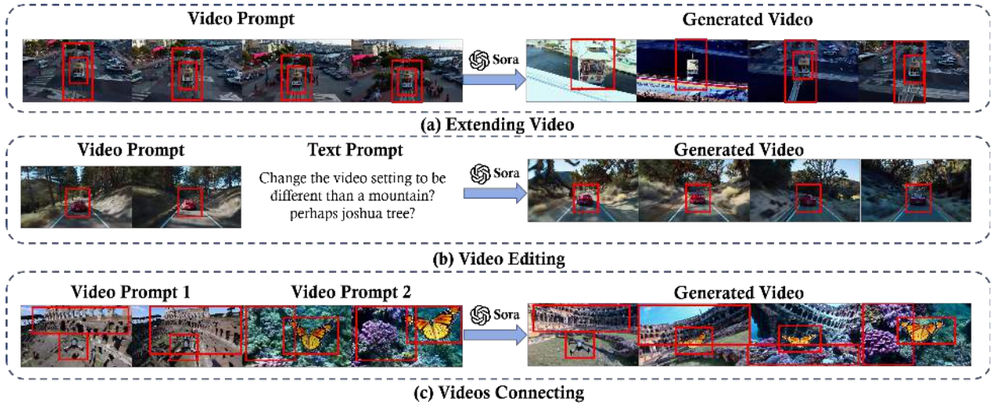


Figure 17: These examples illustrate the video prompt techniques for Sora models: (a) Video Extension, where the model extrapolates the sequence backward the original footage, (b) Video Editing, where specific elements like the setting are transformed as per the text prompt, and (c) Video Connection, where two distinct video prompts are seamlessly blended to create a coherent narrative. Each process is guided by a visual anchor, marked by a red box, ensuring continuity and precision in the generated video content.

nomenon raises critical questions about the reliability and trustworthiness of model outputs, necessitating a comprehensive approach to both evaluate and address the issue. Amount of studies have been dedicated to dissecting the problem of hallucination from various angles. This includes efforts aimed at evaluating the extent and nature of hallucination across different models and scenarios [120, 107, 121, 122]. These evaluations provide invaluable insights into how and why hallucinations occur, laying the groundwork for developing strategies to mitigate their incidence. Concurrently, a significant body of research is focused on devising and implementing methods to reduce hallucinations in these large models [123, 124, 125].

Another vital aspect of trustworthiness is fairness and bias. The critical importance of developing models that do not perpetuate or exacerbate societal biases is a paramount concern. This priority stems from the recognition that biases encoded within these models can reinforce existing social inequities, leading to discriminatory outcomes. Studies in this area, as evidenced by the work of Gallegos et al. [126], Zhang et al. [127], Liang et al. [128], and Friedrich et al. [129], are dedicated to the meticulous identification and rectification of these inherent biases. The goal is to cultivate models that operate fairly, treating all individuals equitably without bias towards race, gender, or other sensitive attributes. This involves not only detecting and mitigating bias in datasets but also designing algorithms that can actively counteract the propagation of such biases [130, 131].

Privacy preservation emerges as another foundational pillar when these models are deployed. In an era where data privacy concerns are escalating, the emphasis on protecting user data has never been more critical. The increasing public awareness and concern over how personal data is handled have prompted more rigorous evaluations of large models. These evaluations focus on the models' capacity to protect user data, ensuring that personal information remains confidential and is not inadvertently disclosed. Research by Mireshghallah et al. [132], Plant et al. [133], and Li et al. [134] exemplify efforts to advance methodologies and technologies that safeguard privacy.

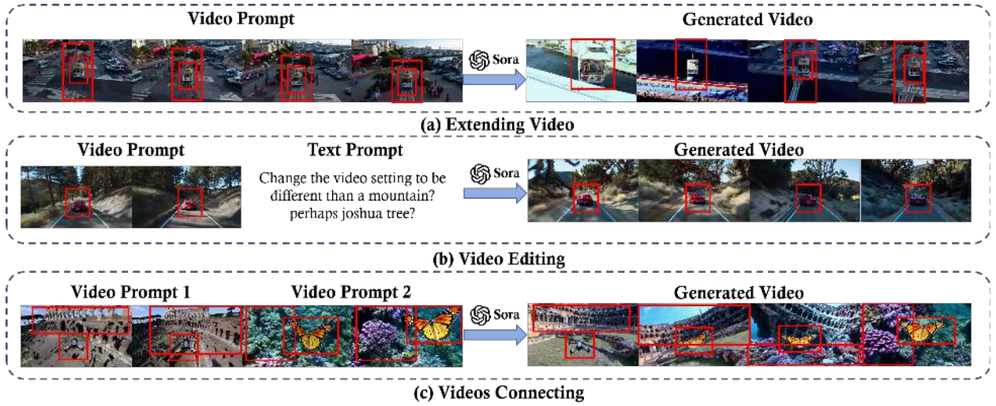


图17：这些示例展示了用于Sora模型的视频提示词技术：(a)视频扩展技术，即模型根据原始视频片段向回推断出后续内容；(b)视频编辑技术，即按照文本提示词对场景等特定元素进行修改；(c)视频拼接技术，即将两个不同的视频提示词无缝融合，从而形成连贯的叙事。每个处理过程都由一个用红色框标出的视觉锚点来引导，以此确保生成的视频内容具备连贯性与精确性。

这类现象引发了人们对模型输出可靠性与可信度的严重质疑，因此需要采取全面的方法来评估并解决这一问题。已有大量研究从不同角度深入剖析幻觉现象的问题，其中包括旨在评估各类模型及不同场景下幻觉现象的规模与性质的探索 [120, 107, 121, 122]。这些评估为我们了解幻觉现象的产生机制与原因提供了宝贵的见解，也为制定减少其发生的策略奠定了基础。与此同时，还有大量研究致力于设计并实施各种方法，以降低这些大型模型中的幻觉现象出现频率 [123, 124, 125]。

可信度的另一个重要方面就是公平性与偏差问题。研发不会延续或加剧社会偏见的人工智能技术模型，具有极其重要的意义。之所以将此视为重中之重，是因为人们意识到这些模型中存在的偏见可能会固化现有的社会不平等现象，进而导致歧视性结果。Gallegos等人 [126]，、Zhang等人 [127]，、 Liang 等人 [128]，以及 Friedrich 等人 [129]，的相关研究都致力于精准识别并纠正这些固有的偏见。其目标在于打造能够公平运行的模型，以无偏见的姿态平等对待所有个体，不论其种族、性别或其他敏感属性如何。这不仅要求检测并减少数据集中的偏见，还要求设计出能够主动抑制此类偏见传播的算法 [130, 131]。

在将这些模型投入实际应用时，隐私保护成为了另一项至关重要的基础支柱。在数据隐私问题日益严重的当下，保护用户数据的重要性前所未有。随着公众对个人数据处理方式的认识和关注不断提升，人们对大型模型的评估标准也愈发严格。这类评估主要着眼于模型保护用户数据的能力，确保个人信息得以保密，不会被无意中泄露。Mireshghallah等人 [132]，、Plant等人 [133]，以及 Li等人 [134] 的研究，正是为推进能够保障隐私的保护方法与技术所做出的努力典范。

3.6.3 Alignment

In addressing these challenges, ensuring the trustworthiness of large models has become one of the primary concerns for researchers [135, 107, 110, 136]. Among the most important technologies is model alignment [136, 137], which refers to the process and goal of ensuring that the behavior and outputs of models are consistent with the intentions and ethical standards of human designers. This concerns the development of technology, its moral responsibilities, and social values. In the domain of LLMs, the method of Reinforcement Learning with Human Feedback (RLHF) [138, 139] has been widely applied for model alignment. This method combines Reinforcement Learning (RL) with direct human feedback, allowing models to better align with human expectations and standards in understanding and performing tasks.

3.6.4 Discussion

From Sora (specifically its technical report), we summarize some insightful findings that potentially offer an informative guideline for future work:

(1) *Integrated Protection of Model and External Security*: As models become more powerful, especially in generating content, ensuring that they are not misused to produce harmful content (such as hate speech [140] and false information [103, 102]) has become a serious challenge. In addition to aligning the model itself, external security protections are equally important. This includes content filtering and review mechanisms, usage permissions and access control, data privacy protection, as well as enhancements in transparency and explainability. For instance, OpenAI now uses a detection classifier to tell whether a given video is generated by Sora [141]. Moreover, a text classifier is deployed to detect the potentially harmful textual input [141].

(2) *Security Challenges of Multimodal Models*: Multimodal models, such as text-to-video models like Sora bring additional complexity to security due to their ability to understand and generate various types of content (text, images, videos, etc.). Multimodal models can produce content in various forms, increasing the ways and scope of misuse and copyright issues. As the content generated by multimodal models is more complex and diverse, traditional methods of content verification and authenticity may no longer be effective. This requires the development of new technologies and methods to identify and filter harmful content generated by these models, increasing the difficulty of regulation and management.

(3) *The Need for Interdisciplinary Collaboration*: Ensuring the safety of models is not just a technical issue but also requires cross-disciplinary cooperation. To address these challenges, experts from various fields such as law [142] and psychology [143] need to work together to develop appropriate norms (e.g., what's the safety and what's unsafe?), policies, and technological solutions. The need for interdisciplinary collaboration significantly increases the complexity of solving these issues.

4 Applications

As video diffusion models, exemplified by Sora, emerge as a forefront technology, their adoption across diverse research fields and industries is rapidly accelerating. The implications of this technology extend far beyond mere video creation, offering transformative potential for tasks ranging from automated content generation to complex decision-making processes. In this section, we delve into a comprehensive examination of the current applications of video diffusion models, highlighting key areas where Sora has not only demonstrated its capabilities but also revolutionized the approach to solving complex problems. We aim to offer a broad perspective for the practical deployment scenarios (see Figure 18).

4.1 Movie

Traditionally, creating cinematic masterpieces has been an arduous and expensive process, often requiring decades of effort, cutting-edge equipment, and substantial financial investments. However, the advent

3.6.3 对齐机制

为应对这些挑战，确保大型模型的可信度已成为研究人员最为关注的问题之一 [135, 107, 110, 136]。其中最重要的技术之一便是模型对齐[136, 137], ，它指的是让模型的行为与输出符合人类设计者的意图及伦理标准的过程与目标。这涉及技术的发展、其道德责任以及社会价值观。在大型语言模型领域，基于人类反馈的强化学习（RLHF）方法 [138, 139] 已被广泛用于模型对齐。该方法将强化学习与直接的人类反馈相结合，使模型在理解任务和执行任务时能更好地契合人类的期望与标准。

3.6.4 讨论

从Sora（尤其是其技术报告）中，我们总结出了一些富有洞察力的研究结果，这些结果或许能为未来的相关工作提供有价值的指导方针。

(1) 模型保护与外部安全的综合防护：随着模型能力的不断提升，尤其是在内容生成方面，如何防止其被滥用来制作有害内容（如仇恨言论 [140]和虚假信息 [103, 102]）已成为一项严峻的挑战。除了对模型本身进行优化外，外部安全防护同样至关重要。这包括内容过滤与审核机制、使用权限与访问控制、数据隐私保护，以及提升透明度与可解释性。例如，OpenAI目前使用检测分类器来判断某个视频是否由Sora生成 [141]。此外，还部署了文本分类器来识别可能具有危害性的文本输入 [141]。

(2) 多模态模型的安全挑战：诸如Sora这类文本转视频模型在内的多模态模型，因其具备理解与生成各类内容（文本、图像、视频等）的能力，给安全防护带来了更多复杂性。这类模型能够以多种形式生成内容，从而增加了滥用行为及版权问题的发生方式和范围。由于多模态模型生成的内容愈发复杂多样，传统的内容验证与真实性检测方法可能已不再有效。这就需要开发新的技术手段来识别并过滤这些模型生成的有害内容，进而提升了监管与管理的难度。

(3) 跨学科协作的必要性：确保模型安全不仅是一个技术问题，还需要跨学科的合作。为应对这些挑战，来自法律 [142]和心理学 [143] 等各个领域的专家需要通力协作，共同制定相应的规范（例如什么属于安全行为，什么属于危险行为？）、政策以及技术解决方案。跨学科协作的需求大大增加了解决这些问题的复杂性。

4种应用场景

作为以Sora为代表的视频扩散模型，作为一种前沿技术，正迅速被广泛应用于各种研究领域和产业中。这项技术的影响力远不止于简单的视频创作，它为从自动化内容生成到复杂决策流程等各类任务带来了颠覆性的潜力。在本节中，我们将全面探讨当前视频扩散模型的应用情况，重点介绍Sora 不仅在诸多领域展现出强大能力，更彻底改变了解决复杂问题方式的那些关键领域。我们旨在为实际应用场景提供更为广阔的视角（见图18）。

4.1 电影

传统上，打造电影级杰作一直是一项既艰巨又昂贵的工作，往往需要数十年的努力、最先进的设备以及巨额的资金投入。然而，随着DiT模型的出现

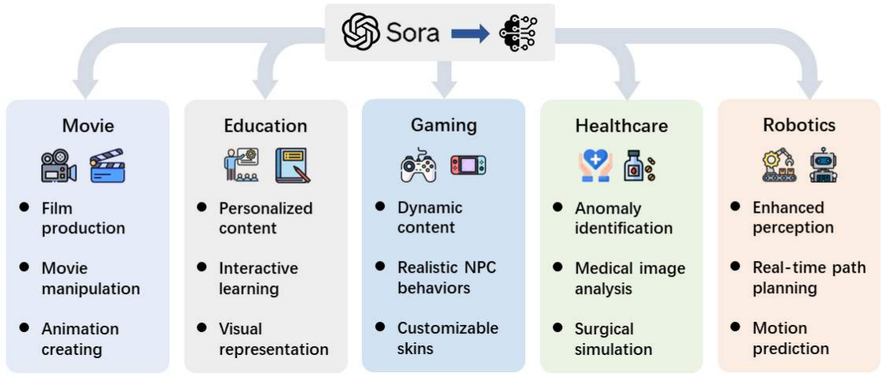


Figure 18: Applications of Sora.

of advanced video generation technologies heralds a new era in film-making, one where the dream of autonomously producing movies from simple text inputs is becoming a reality. Researchers have ventured into the realm of movie generation by extending video generation models into creating movies. MovieFactory [144] applies diffusion models to generate film-style videos from elaborate scripts produced by ChatGPT [100], representing a significant leap forward. In the follow-up, MobileVidFactory [145] can automatically generate vertical mobile videos with only simple texts provided by users. Vlogger [146] makes it feasible for users to compose a minute-long vlog. These developments, epitomized by Sora's ability to generate captivating movie content effortlessly, mark a pivotal moment in the democratization of movie production. They offer a glimpse into a future where anyone can be a filmmaker, significantly lowering the barriers to entry in the film industry and introducing a novel dimension to movie production that blends traditional storytelling with AI-driven creativity. The implications of these technologies extend beyond simplification. They promise to reshape the landscape of film production, making it more accessible and versatile in the face of evolving viewer preferences and distribution channels.

4.2 Education

The landscape of educational content has long been dominated by static resources, which, despite their value, often fall short of catering to the diverse needs and learning styles of today's students. Video diffusion models stand at the forefront of an educational revolution, offering unprecedented opportunities to customize and animate educational materials in ways that significantly enhance learner engagement and understanding. These advanced technologies enable educators to transform text descriptions or curriculum outlines into dynamic, engaging video content tailored to the specific style, and interests of individual learners [147, 148, 149, 150]. Moreover, image-to-video editing techniques [151, 152, 153] present innovative avenues for converting static educational assets into interactive videos, thereby supporting a range of learning preferences and potentially increasing student engagement. By integrating these models into educational content creation, educators can produce videos on a myriad of subjects, making complex concepts more accessible and captivating for students. The use of Sora in revolutionizing the educational domain exemplifies the transformative potential of these technologies. This shift towards personalized, dynamic educational content heralds a new era in education.

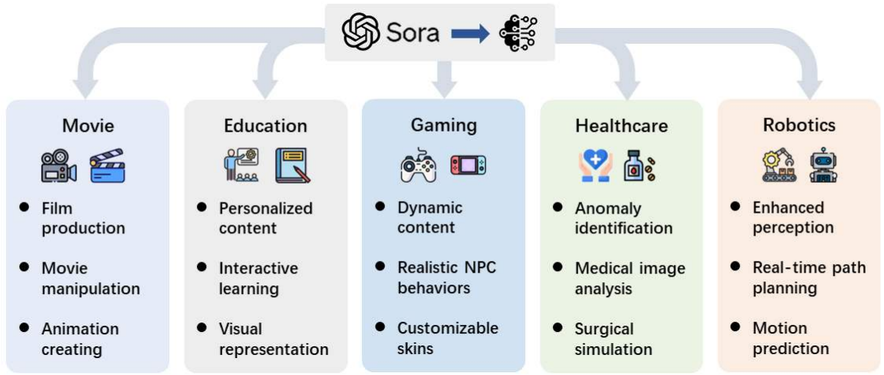


图18：Sora的应用场景。

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4.2 教育领域

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4.3 Gaming

The gaming industry constantly seeks ways to push the boundaries of realism and immersion, yet traditional game development often grapples with the limitations of pre-rendered environments and scripted events. The generation of dynamic, high-fidelity video content and realistic sound by diffusion models effects in real-time, promise to overcome existing constraints, offering developers the tools to create evolving game environments that respond organically to player actions and game events [154, 155]. This could include generating changing weather conditions, transforming landscapes, or even creating entirely new settings on the fly, making game worlds more immersive and responsive. Some methods [156, 157] also synthesize realistic impact sounds from video inputs, enhancing game audio experiences. With the integration of `Sora` within the gaming domain, unparalleled immersive experiences that captivate and engage players can be created. How games are developed, played, and experienced will be innovated, as well as opening new possibilities for storytelling, interaction, and immersion.

4.4 Healthcare

Despite generative capabilities, video diffusion models excel in understanding and generating complex video sequences, making them particularly suited for identifying dynamic anomalies within the body, such as early cellular apoptosis [158], skin lesion progression [159], and irregular human movements [160], which are crucial for early disease detection and intervention strategies. Additionally, models like MedSegDiff-V2 [161] and [162] leverage the power of transformers to segment medical images with unprecedented precision, enabling clinicians to pinpoint areas of interest across various imaging modalities with enhanced accuracy. The integration of `Sora` into clinical practice promises not only to refine diagnostic processes but also to personalize patient care, offering tailored treatment plans based on precise medical imaging analysis. However, this technological integration comes with its own set of challenges, including the need for robust data privacy measures and addressing ethical considerations in healthcare.

4.5 Robotics

Video diffusion models now play important roles in robotics, showing a new era where robots can generate and interpret complex video sequences for enhanced perception [163, 164] and decision-making [165, 166, 167]. These models unlock new capabilities in robots, enabling them to interact with their environment and execute tasks with unprecedented complexity and precision. The introduction of web-scale diffusion models to robotics [163] showcases the potential for leveraging large-scale models to enhance robotic vision and understanding. Latent diffusion models are employed for language-instructed video prediction [168], allowing robots to understand and execute tasks by predicting the outcome of actions in video format. Furthermore, the reliance on simulated environments for robotics research has been innovatively addressed by video diffusion models capable of creating highly realistic video sequences [169, 170]. This enables the generation of diverse training scenarios for robots, mitigating the limitations imposed by the scarcity of real-world data. We believe, the integration of technologies like `Sora` into the robotics field holds the promise of groundbreaking developments. By harnessing the power of `Sora`, the future of robotics is poised for unprecedented advancements, where robots can seamlessly navigate and interact with their environments.

5 Discussion

`Sora` shows a remarkable talent for precisely understanding and implementing complex instructions from humans. This model excels at creating detailed videos with various characters, all set within elaborately crafted settings. A particularly impressive attribute of `Sora` is its ability to produce videos up to one minute in length while ensuring consistent and engaging storytelling. This marks a significant improvement over previous attempts that focused on shorter video pieces, as `Sora`'s extended sequences exhibit a clear narrative flow and maintain visual consistency from start to finish. Furthermore, `Sora` distinguishes itself

4.3 游戏领域

The gaming industry constantly seeks ways to push the boundaries of realism and immersion, yet traditional game development often grapples with the limitations of pre-rendered environments and scripted events. The generation of dynamic, high-fidelity video content and realistic sound by diffusion models effects in real-time, promise to overcome existing constraints, offering developers the tools to create evolving game environments that respond organically to player actions and game events [154, 155]. This could include generating changing weather conditions, transforming landscapes, or even creating entirely new settings on the fly, making game worlds more immersive and responsive. Some methods [156, 157] also synthesize realistic impact sounds from video inputs, enhancing game audio experiences. With the integration of `Sora` within the gaming domain, unparalleled immersive experiences that captivate and engage players can be created. How games are developed, played, and experienced will be innovated, as well as opening new possibilities for storytelling, interaction, and immersion.

4.4 医疗保健

尽管具备生成能力，视频扩散模型在理解和生成复杂的视频序列方面表现极为出色，这使其特别适合用于识别人体内的动态异常，比如早期的细胞凋亡现象、 [158], 皮肤病变的进展 [159], 以及不规则的人体动作 [160], ，而这些都是实现疾病早期检测和制定干预策略的关键。此外，MedSegDiff-V2 [161] 和 [162] 这类模型还借助变换器的技术优势，以前所未有的精度对医学图像进行分割，从而使临床医生能够更精准地定位各种影像检查中的关键区域。将Sora引入临床实践不仅有望优化诊断流程，还能实现个性化医疗护理，通过精确的医学影像分析为患者提供定制化的治疗方案。不过，这种技术整合也带来了一系列挑战，包括需要采取强有力的数据隐私保护措施，以及妥善处理医疗领域中的伦理问题。

4.5 机器人技术

Video diffusion models now play important roles in robotics, showing a new era where robots can generate and interpret complex video sequences for enhanced perception [163, 164] and decision-making [165, 166, 167]. These models unlock new capabilities in robots, enabling them to interact with their environment and execute tasks with unprecedented complexity and precision. The introduction of web-scale diffusion models to robotics [163] showcases the potential for leveraging large-scale models to enhance robotic vision and understanding. Latent diffusion models are employed for language-instructed video prediction [168], allowing robots to understand and execute tasks by predicting the outcome of actions in video format. Furthermore, the reliance on simulated environments for robotics research has been innovatively addressed by video diffusion models capable of creating highly realistic video sequences [169, 170]. This enables the generation of diverse training scenarios for robots, mitigating the limitations imposed by the scarcity of real-world data. We believe, the integration of technologies like `Sora` into the robotics field holds the promise of groundbreaking developments. By harnessing the power of `Sora`, the future of robotics is poised for unprecedented advancements, where robots can seamlessly navigate and interact with their environments.

5 讨论

Sora在精准理解并执行人类给出的复杂指令方面展现出非凡的能力。该模型擅长制作包含多种角色的详细视频，且所有场景都经过精心设计。Sora最令人赞叹的特点之一，便是它能够生成时长达一分钟的视频，同时还能保证叙事连贯且富有吸引力。相比以往那些仅能处理较短视频片段的尝试，这一进步极为显著，因为Sora生成的较长序列具备清晰的叙事脉络，并且从开头到结尾都能保持视觉一致性。此外，Sora还凭借另一项优势脱颖而出。

by generating longer video sequences that capture complex movements and interactions, advancing past the restrictions of earlier models that could only handle short clips and basic images. This advancement signifies a major step forward in AI-powered creative tools, enabling users to transform written stories into vivid videos with a level of detail and sophistication that was previously unattainable.

5.1 Limitations

Challenges in Physical Realism. Sora, as a simulation platform, exhibits a range of limitations that undermine its effectiveness in accurately depicting complex scenarios. Most important is its inconsistent handling of physical principles within complex scenes, leading to a failure in accurately copying specific examples of cause and effect. For instance, consuming a portion of a cookie might not result in a corresponding bite mark, illustrating the system's occasional departure from physical plausibility. This issue extends to the simulation of motion, where Sora generates movements that challenge realistic physical modeling, such as unnatural transformations of objects or the incorrect simulation of rigid structures like chairs, leading to unrealistic physical interactions. The challenge further increases when simulating complex interactions among objects and characters, occasionally producing outcomes that lean towards the humorous.

Spatial and Temporal Complexities. Sora occasionally misunderstands instructions related to the placement or arrangement of objects and characters within a given prompt, leading to confusion about directions (e.g., confusing left for right). Additionally, it faces challenges in maintaining the temporal accuracy of events, particularly when it comes to adhering to designated camera movements or sequences. This can result in deviating from the intended temporal flow of scenes. In complex scenarios that involve a multitude of characters or elements, Sora has a tendency to insert irrelevant animals or people. Such additions can significantly change the originally envisioned composition and atmosphere of the scene, moving away from the planned narrative or visual layout. This issue not only affects the model's ability to accurately recreate specific scenes or narratives but also impacts its reliability in generating content that closely aligns with the user's expectations and the coherence of the generated output.

Limitations in Human-computer Interaction (HCI). Sora, while showing potential in the video generation domain, faces significant limitations in HCI. These limitations are primarily evident in the coherence and efficiency of user-system interactions, especially when making detailed modifications or optimizations to generated content. For instance, users might find it difficult to precisely specify or adjust the presentation of specific elements within a video, such as action details and scene transitions. Additionally, Sora's limitations in understanding complex language instructions or capturing subtle semantic differences could result in video content that does not fully meet user expectations or needs. These shortcomings restrict Sora's potential in video editing and enhancement, also impacting the overall satisfaction of the user experience.

Usage Limitation. Regarding usage limitations, OpenAI has not yet set a specific release date for public access to Sora, emphasizing a cautious approach towards safety and readiness before broad deployment. This indicates that further improvements and testing in areas such as security, privacy protection, and content review may still be necessary for Sora. Moreover, at present, Sora can only generate videos up to one minute in length, and according to published cases, most generated videos are only a few dozen seconds long. This limitation restricts its use in applications requiring longer content display, such as detailed instructional videos or in-depth storytelling. This limitation reduces Sora's flexibility in the content creation.

5.2 Opportunities

Academy. (1) The introduction of Sora by OpenAI marks a strategic shift towards encouraging the broader AI community to delve deeper into the exploration of text-to-video models, leveraging both diffusion and transformer technologies. This initiative aims to redirect the focus toward the potential of creating highly sophisticated and nuanced video content directly from textual descriptions, a frontier that promises to rev-

它能够生成捕捉复杂动作与互动的更长视频序列，从而突破了早期模型只能处理短片段和简单图像的限制。这一进展标志着人工智能驱动的创意工具领域迈出了重要一步，让用户能够将文字故事转化为细节丰富、精致度极高的生动视频，实现了前所未有的效果。

5.1 局限性

物理真实性的挑战。作为一款模拟平台，Sora存在诸多限制，这些限制削弱了其在精准呈现复杂场景方面的效果。其中最突出的是它在处理复杂场景中的物理原理时表现不一致，从而导致无法准确还原特定的因果关系实例。例如，吃掉一块饼干可能并不会留下相应的咬痕，这恰恰体现了该系统有时会偏离物理合理性。这一问题在运动模拟方面也同样存在，Sora生成的动作往往难以符合真实的物理建模规律，比如物体会出现不自然的变形，或者椅子这类刚性结构的模拟出现错误，进而引发不真实的物理交互。而在模拟物体与角色之间的复杂互动时，这一挑战更为严峻，系统偶尔还会产生一些近乎幽默的结局。

空间与时间层面的复杂性。 Sora偶尔会误解提示词中关于物体和角色位置或排列的指令，从而导致方向混淆（例如将左误认为右）。此外，它在保持事件的时间准确性方面也存在困难，尤其是在遵循指定的摄像机移动或序列时。这可能会使场景的时间流程偏离预期。在涉及众多角色或元素的复杂场景中，Sora还倾向于添加无关的动物或人物。这类添加物会显著改变场景原本设想的构图与氛围，使内容偏离既定的叙事或视觉布局。这一问题不仅会影响模型准确再现特定场景或叙事的能力，还会降低其生成符合用户期望的内容的可靠性，以及输出结果的连贯性。

人机交互方面的局限性。Sora虽然在视频生成领域展现出巨大潜力，但在人机交互方面存在显著局限。这些局限主要体现在用户与系统交互的连贯性及效率上，尤其是在对生成内容进行详细修改或优化时更为明显。例如，用户可能难以精准指定或调整视频中特定元素的呈现方式，比如动作细节和场景切换等。此外，Sora在理解复杂的语言指令或捕捉细微的语义差异方面存在不足，这可能导致生成的视频内容无法完全满足用户的期望与需求。这些缺陷限制了Sora在视频编辑与优化方面的应用潜力，同时也影响了整体用户体验的满意度。

使用限制。在使用限制方面，OpenAI尚未确定让公众使用Sora的具体发布日期，强调在大规模部署之前需谨慎处理安全与准备问题。这表明Sora可能仍需要在安全性、隐私保护以及内容审核等领域进行进一步的优化和测试。此外，目前Sora只能生成长度不超过一分钟的视频，从已公开的案例来看，大多数生成的视频时长也只有几十秒。这一限制使其难以应用于需要更长内容展示的场景，比如详细的操作教程或深入的故事讲述。该限制也降低了Sora在内容创作方面的灵活性。

5.2 发展机遇

学术界。(1) OpenAI推出的Sora标志着一项战略性的转变，旨在鼓励更广泛的人工智能社区深入探索文本转视频模型，同时运用扩散模型与Transformer技术。这一举措的目的是将研究重点转向直接从文字描述创建高度复杂且富有细节的视频内容的潜力上，而这正是有望带来革新的前沿领域。

olutionize content creation, storytelling, and information sharing. (2) The innovative approach of training Sora on data at its native size, as opposed to the traditional methods of resizing or cropping, serves as a groundbreaking inspiration for the academic community. It opens up new pathways by highlighting the benefits of utilizing unmodified datasets, which leads to the creation of more advanced generative models.

Industry. (1) The current capabilities of Sora signal a promising path for the advancement of video simulation technologies, highlighting the potential to significantly enhance realism within both physical and digital areas. The prospect of Sora enabling the creation of highly realistic environments through textual descriptions presents a promising future for content creation. This potential extends to revolutionizing game development, offering a glimpse into a future where immersive-generated worlds can be crafted with unprecedented ease and accuracy. (2) Companies may leverage Sora to produce advertising videos that swiftly adapt to market changes and create customized marketing content. This not only reduces production costs but also enhances the appeal and effectiveness of advertisements. The ability of Sora to generate highly realistic video content from textual descriptions alone could revolutionize how brands engage with their audience, allowing for the creation of immersive and compelling videos that capture the essence of their products or services in unprecedented ways.

Society. (1) While the prospect of utilizing text-to-video technology to replace traditional filmmaking remains distant, Sora and similar platforms hold transformative potential for content creation on social media. The constraints of current video lengths do not diminish the impact these tools can have in making high-quality video production accessible to everyone, enabling individuals to produce compelling content without the need for expensive equipment. It represents a significant shift towards empowering content creators across platforms like TikTok and Reels, bringing in a new age of creativity and engagement. (2) Screenwriters and creative professionals can use Sora to transform written scripts into videos, assisting them in better showing and sharing their creative concepts, and even in producing short films and animations. The ability to create detailed, vivid videos from scripts can fundamentally change the pre-production process of filmmaking and animation, offering a glimpse into how future storytellers might pitch, develop, and refine their narratives. This technology opens up possibilities for a more dynamic and interactive form of script development, where ideas can be visualized and assessed in real time, providing a powerful tool for creativity and collaboration. (3) Journalists and news organizations can also utilize Sora to quickly generate news reports or explanatory videos, making the news content more vivid and engaging. This can significantly increase the coverage and audience engagement of news reports. By providing a tool that can simulate realistic environments and scenarios, Sora offers a powerful solution for visual storytelling, enabling journalists to convey complex stories through engaging videos that were previously difficult or expensive to produce. In summary, Sora's potential to revolutionize content creation across marketing, journalism, and entertainment is immense.

6 Conclusion

We present a comprehensive review of Sora to help developers and researchers study the capabilities and related works of Sora. The review is based on our survey of published technical reports and reverse engineering based on existing literature. We will continue to update the paper when Sora's API is available and further details about Sora are revealed. We hope that this review paper will prove a valuable resource for the open-source research community and lay a foundation for the community to jointly develop an open-source version of Sora in the near future to democratize video auto-creation in the era of AIGC. To achieve this goal, we invite discussions, suggestions, and collaborations on all fronts.

实现内容创作、故事讲述与信息共享的自动化。(2) 相比于传统的调整尺寸或裁剪方法，直接以原始规模的数据来训练Sora这一创新方式，为学术界带来了极具开创性的启示。它通过凸显使用未经修改的数据集的优势，开辟了新的发展路径，进而助力更先进的生成式模型的诞生。

行业现状。(1) 目前Sora所具备的能力为视频模拟技术的发展指明了一条充满希望的道路，它展现出在实体世界与数字领域大幅提升真实感的巨大潜力。通过文字描述即可创建高度逼真环境这一前景，为内容创作带来了光明的未来。这种潜力还足以革新游戏开发行业，让人们预见一个能够以前所未有的便捷性与精准度打造沉浸式生成世界的未来。(2) 各企业可以利用Sora制作能够快速响应市场变化的广告视频，并创建定制化的营销内容。这不仅能够降低制作成本，还能提升广告的吸引力与效果。Sora仅需依据文字描述就能生成高度逼真的视频内容，这一能力有望彻底改变品牌与受众的互动方式，帮助它们以独特的方式制作出富有沉浸感且极具感染力的视频，从而将产品或服务的精髓展现得淋漓尽致。

Society. (1) While the prospect of utilizing text-to-video technology to replace traditional filmmaking remains distant, Sora and similar platforms hold transformative potential for content creation on social media. The constraints of current video lengths do not diminish the impact these tools can have in making high-quality video production accessible to everyone, enabling individuals to produce compelling content without the need for expensive equipment. It represents a significant shift towards empowering content creators across platforms like TikTok and Reels, bringing in a new age of creativity and engagement. (2) Screenwriters and creative professionals can use Sora to transform written scripts into videos, assisting them in better showing and sharing their creative concepts, and even in producing short films and animations. The ability to create detailed, vivid videos from scripts can fundamentally change the pre-production process of filmmaking and animation, offering a glimpse into how future storytellers might pitch, develop, and refine their narratives. This technology opens up possibilities for a more dynamic and interactive form of script development, where ideas can be visualized and assessed in real time, providing a powerful tool for creativity and collaboration. (3) Journalists and news organizations can also utilize Sora to quickly generate news reports or explanatory videos, making the news content more vivid and engaging. This can significantly increase the coverage and audience engagement of news reports. By providing a tool that can simulate realistic environments and scenarios, Sora offers a powerful solution for visual storytelling, enabling journalists to convey complex stories through engaging videos that were previously difficult or expensive to produce. In summary, Sora's potential to revolutionize content creation across marketing, journalism, and entertainment is immense.

6 结论

我们撰写了这篇关于Sora的全面综述，旨在帮助开发人员和研究人员了解其功能及相关研究成果。该综述基于我们对已发表技术报告的梳理，以及通过对现有文献的逆向分析而形成。一旦Sora的API正式开放且有关它的更多细节被披露，我们将持续更新这篇论文。我们希望这篇综述能为开源研究社区提供宝贵的资源，并为社区在不久的将来共同开发出Sora的开源版本奠定基础，从而在AIGC时代推动视频自动化创作技术的普及。为实现这一目标，我们欢迎各方提出讨论意见、建议以及开展合作。

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A Related Works

We show some related works about the video generation tasks in Table 1.

Table 1: Summary of Video Generation.

Model name	Year	Backbone	Task	Group
Imagen Video[29]	2022	Diffusion	Generation	Google
Pix2Seq-D[171]	2022	Diffusion	Segmentation	Google Deepmind
FDM[172]	2022	Diffusion	Prediction	UBC
MaskViT[173]	2022	Masked Vision Models	Prediction	Stanford, Salesforce
CogVideo[174]	2022	Auto-regressive	Generation	THU
Make-a-video[175]	2022	Diffusion	Generation	Meta
MagicVideo[176]	2022	Diffusion	Generation	ByteDance
TATS[177]	2022	Auto-regressive	Generation	University of Maryland, Meta
Phenaki[178]	2022	Masked Vision Models	Generation	Google Brain
Gen-1[179]	2023	Diffusion	Generation, Editing	RunwayML
LFDM[151]	2023	Diffusion	Generation	PSU, UCSD
Text2video-Zero[180]	2023	Diffusion	Generation	Picsart
Video Fusion[181]	2023	Diffusion	Generation	USAC, Alibaba
PYoCo[34]	2023	Diffusion	Generation	Nvidia
Video LDM[36]	2023	Diffusion	Generation	University of Maryland, Nvidia
RIN[182]	2023	Diffusion	Generation	Google Brain
LVD[183]	2023	Diffusion	Generation	UCB
Dreamix[184]	2023	Diffusion	Editing	Google
MagicEdit[185]	2023	Diffusion	Editing	ByteDance
Control-A-Video[186]	2023	Diffusion	Editing	Sun Yat-Sen University
StableVideo[187]	2023	Diffusion	Editing	ZJU, MSRA
Tune-A-Video[89]	2023	Diffusion	Editing	NUS
Rerender-A-Video[188]	2023	Diffusion	Editing	NTU
Pix2Video[189]	2023	Diffusion	Editing	Adobe, UCL
InstructVid2Vid[190]	2023	Diffusion	Editing	ZJU
DiffAct[191]	2023	Diffusion	Action Detection	University of Sydney
DiffPose[192]	2023	Diffusion	Pose Estimation	Jilin University
MAGVIT[193]	2023	Masked Vision Models	Generation	Google
AnimateDiff[149]	2023	Diffusion	Generation	CUHK
MAGVIT V2[47]	2023	Masked Vision Models	Generation	Google
Generative Dynamics[194]	2023	Diffusion	Generation	Google
VideoCrafter[92]	2023	Diffusion	Generation	Tencent
Zeroscope[195]	2023	-	Generation	EasyWithAI
ModelScope	2023	-	Generation	Damo
Gen-2[23]	2023	-	Generation	RunwayML
Pika[22]	2023	-	Generation	Pika Labs
Emu Video[196]	2023	Diffusion	Generation	Meta
PixelDance[197]	2023	Diffusion	Generation	ByteDance
Stable Video Diffusion[27]	2023	Diffusion	Generation	Stability AI
W.A.L.T[65]	2023	Diffusion	Generation	Stanford, Google
Fairy[198]	2023	Diffusion	Generation, Editing	Meta
VideoPoet[199]	2023	Auto-regressive	Generation, Editing	Google
GenTron [64]	2023	Diffusion	Generation	HKU & Meta
LGVI[200]	2024	Diffusion	Editing	PKU, NTU
Lumiere[201]	2024	Diffusion	Generation	Google
Sora[3]	2024	Diffusion	Generation, Editing	OpenAI

相关研究

我们在表 1中列举了一些与视频生成任务相关的现有研究。

表1：视频生成技术概述。

模型名称	年份	通用架构	Task	分组
Imagen Video[29]	2022	扩散模型	生成	谷歌
Pix2Seq-D[171]	2022	扩散模型	分割	谷歌Deepmind
FDM[172]	2022	扩散模型	预测	UBC
MaskViT[173]	2022	掩码视觉模型	预测	斯坦福大学与Salesforce公司
CogVideo[174]	2022	自回归	生成	THU
Make-a-video[175]	2022	扩散模型	生成	Meta
MagicVideo[176]	2022	扩散模型	生成	字节跳动
TATS[177]	2022	自回归	生成	马里兰大学与Meta公司
菲纳基[178]	2022	掩码视觉模型	生成	谷歌大脑
Gen-1[179]	2023	扩散模型	生成与编辑	RunwayML
LFDM[151]	2023	扩散模型	生成	宾夕法尼亚州立大学，加州大学圣地亚哥分校
Text2video-Zero[180]	2023	扩散模型	生成	Picsart软件
视频融合技术[181]	2023	扩散模型	生成	USAC公司与阿里巴巴集团
PYoCo[34]	2023	扩散模型	生成	英伟达
视频LDM算法[36]	2023	扩散模型	生成	马里兰大学与英伟达
RIN[182]	2023	扩散模型	生成	谷歌大脑
LVD[183]	2023	扩散模型	生成	UCB
Dreamix[184]	2023	扩散模型	编辑	谷歌
MagicEdit[185]	2023	扩散模型	编辑	字节跳动
Control-A-Video[186]	2023	扩散模型	编辑	中山大学
StableVideo[187]	2023	扩散模型	编辑	浙江大学与微软亚洲研究院
Tune-A-Video[89]	2023	扩散模型	编辑	NUS
Rerender-A-Video[188]	2023	扩散模型	编辑	NTU
Pix2Video[189]	2023	扩散模型	编辑	Adobe公司、伦敦大学学院
InstructVid2Vid[190]	2023	扩散模型	编辑	ZJU
DiffAct[191]	2023	扩散模型	动作检测	悉尼大学
DiffPose[192]	2023	扩散模型	姿态估计	吉林大学
MAGVIT[193]	2023	掩码视觉模型	生成	谷歌
AnimateDiff[149]	2023	扩散模型	生成	CUHK
MAGVIT V2[47]	2023	掩码视觉模型	生成	谷歌
生成动力学[194]	2023	扩散模型	生成	谷歌
VideoCrafter[92]	2023	扩散模型	生成	腾讯
Zeroscope[195]	2023	-	生成	EasyWithAI
ModelScope	2023	-	生成	Damo
Gen-2[23]	2023	-	生成	RunwayML
Pika[22]	2023	-	生成	Pika Labs
Emu Video[196]	2023	扩散模型	生成	Meta
PixelDance[197]	2023	扩散模型	生成	字节跳动
稳定视频扩散模型[27]	2023	扩散模型	生成	Stability AI公司
W.A.L.T[65]	2023	扩散模型	生成	斯坦福大学与谷歌
Fairy[198]	2023	扩散模型	生成与编辑	Meta
VideoPoet[199]	2023	自回归	生成与编辑	谷歌
GenTron [64]	2023	扩散模型	生成	香港大学与Meta公司
LGVI[200]	2024	扩散模型	编辑	北京大学与新加坡国立大学
Lumiere[201]	2024	扩散模型	生成	谷歌
Sora[3]	2024	扩散模型	生成与编辑	OpenAI